WILMINGTON UNIVERSITY

COLLEGE OF TECHNOLOGY

SEC 210 – Principles and Practice of Information Security

Tyler Sanderson

OBJECTIVES: Perform **Lab 3.2C Active Stack Fingerprinting Using NMap.** This is found in the **Lab Manual** on pages **80 - 81**

INSTRUCTIONS: Carefully follow the instructions on each page. You will submit numerous screenshots. Search for INSERT or SCREENSHOT in uppercase to find out the exact pages. There are questions to answer. Search for QUESTION in uppercase to find the exact pages.

NOTE: You do not need to reference the Lab Manual. This lab document is self-sufficient.

PREREQUISITES:

- "How to use the Snipping Tool on Windows 10" video at https://www.youtube.com/watch?v=ayi5-7QPNcA
- "How to Take a Screenshot on a Mac" video at https://www.youtube.com/watch?v=h3RkDPfphSQ
- How to access SEC 210 vApp.docx

COMMENTS: I do NOT want a screenshot of your entire VM desktop window when only a portion is needed to demonstrate that you have followed the instructions. For examples, see the screenshots in this document.

Ensure that you complete the entire lab. You will know when you see "THIS IS THE LAST PAGE OF THE LAB".

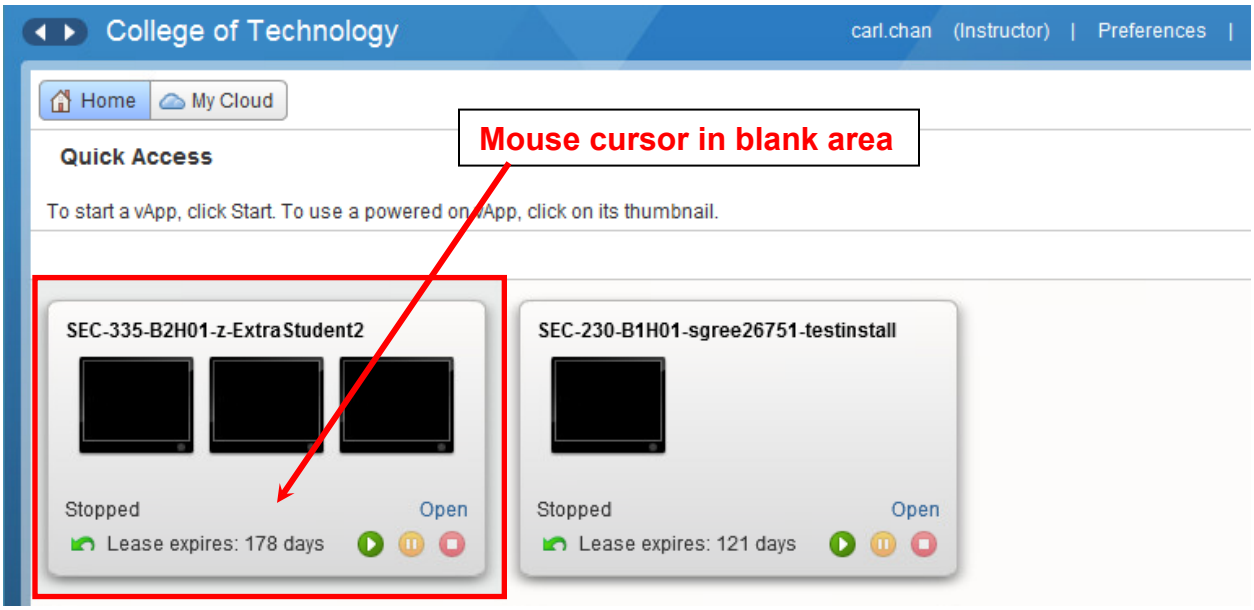
In this lab, you will perform some simple tasks using the **zenmap** utility. To provide more background about this lab and to learn more about other capabilities, watch the video at

https://www.youtube.com/watch?v=o1u69KvSFmQ

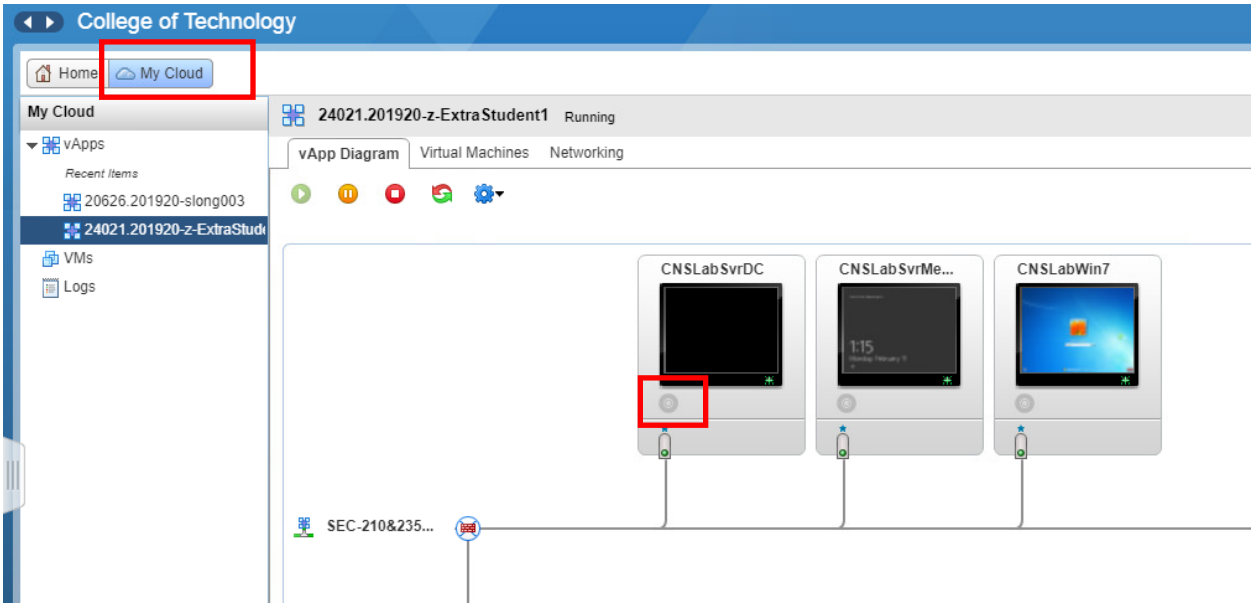
Network Scanning and Reconnaissance with NMAP and ZenMAP (12:08)

First, we will ensure that no DVD is inserted into the **CNSLabSvrDC** VM.

- Place your mouse cursor on a blank area of your SEC 210 vApp, not on one of the VM icons
- **Double-click** on the vApp

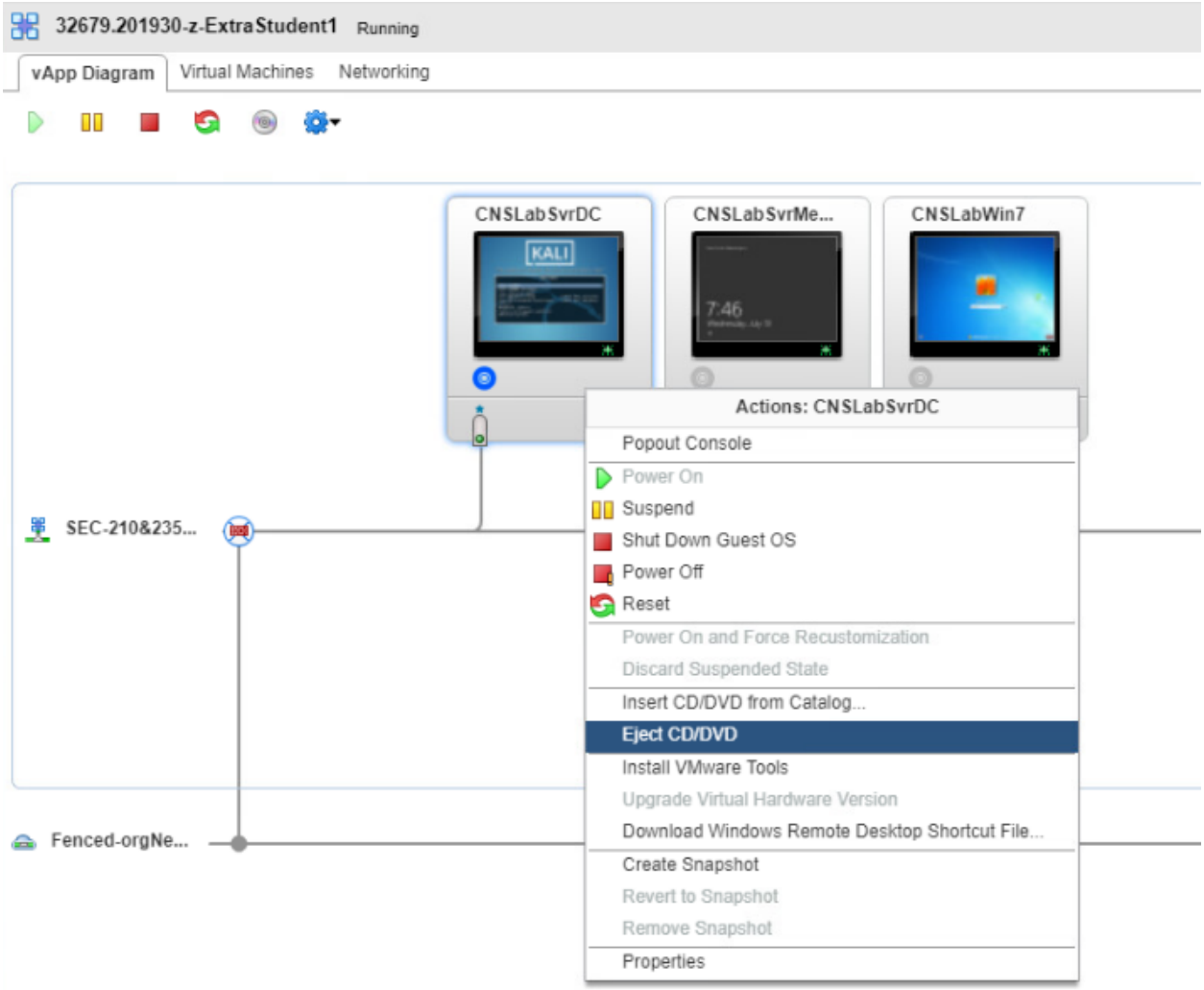


- vApp Diagram folder tab window appears. The **My Cloud folder** tab is now highlighted
- Look at the **DVD** button on the **CNSLabSrvMember** VM. It is the left-most VM
- If it is greyed out as shown, then go directly to **Disable the Firewall on CNSLabSrvMember**
- If is is blue, then go to **Ejecting the DVD**

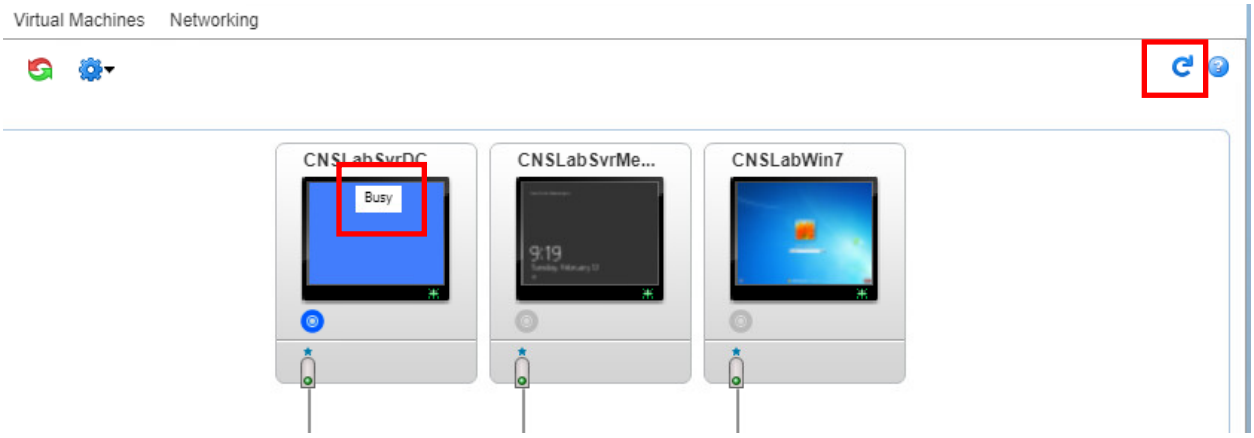


Ejecting the DVD

- Hover mouse cursor over **CNSLabSvrDC**, the left-most VM
- NOTE: The circle is blue out indicating that a DVD has been inserted
- Right-click mouse, pop-up appears
- Select **Eject CD/DVD**



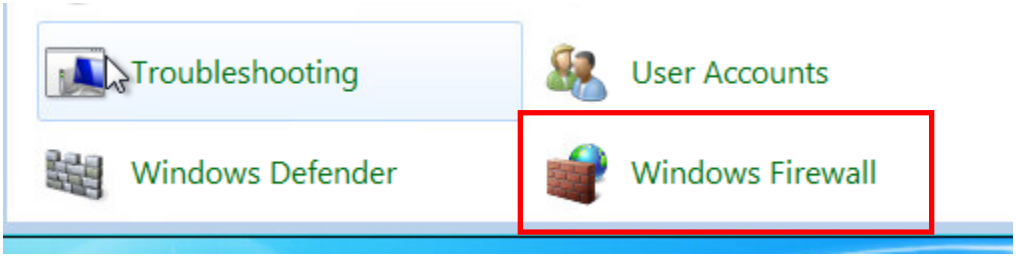
- **Busy** appears in the **CNSLabSvrDC** icon
- Wait 5 minutes and then click on **Refresh** icon. WARNING: Do not click too early.
- The button should now be grey



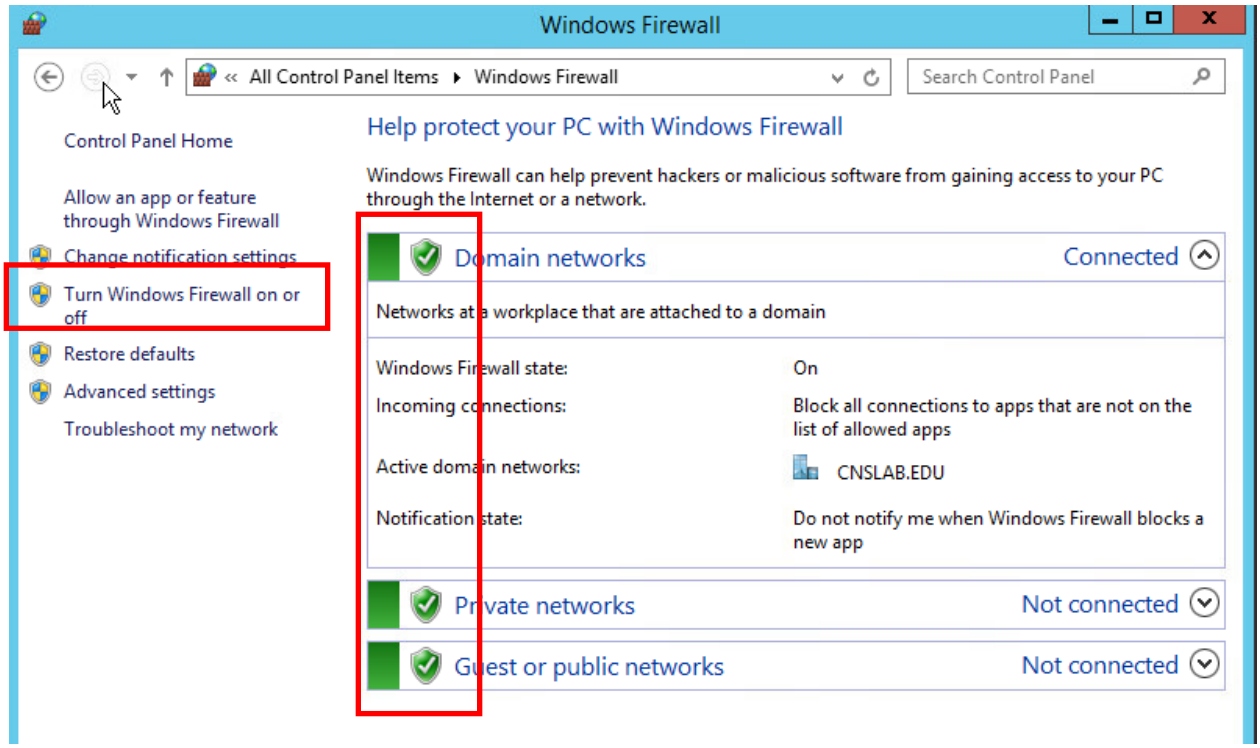
Disable the Firewall on CNSLabSrvMember

Next, we must disable the **Windows Firewall** on **CNSLabSrvMember**

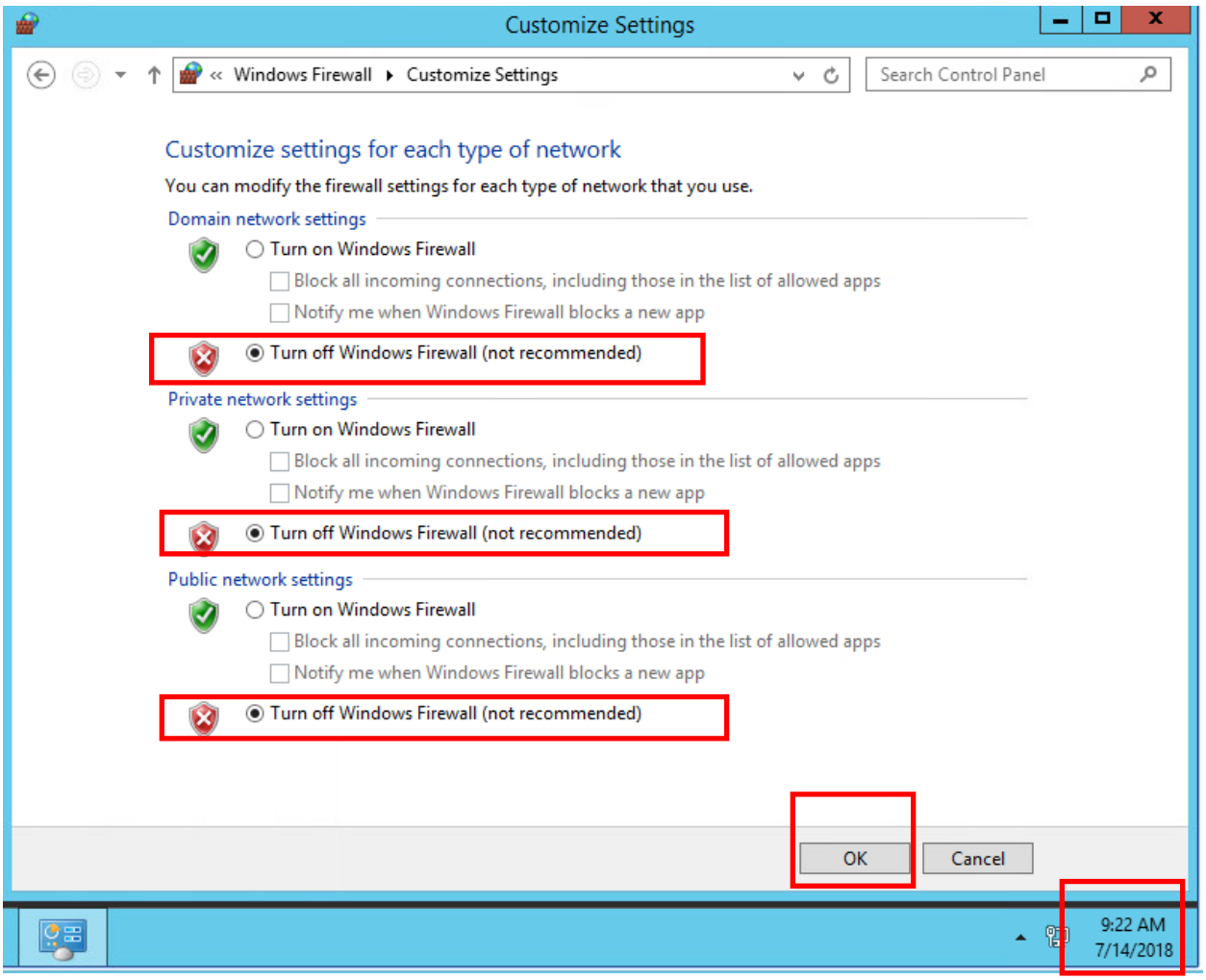
- Login to the **CNSLabSrvMember** VM. The password is "**cnslab**"
- You may need to resize the VM to expose the taskbar at the bottom of the window
- Open **Control Panel**
- Scroll down and click on the "**Windows Firewall**" icon



- The **Windows Filewall** window appears
- All status bars show as green
- Click on "**Turn Windows Firewal on or off**".

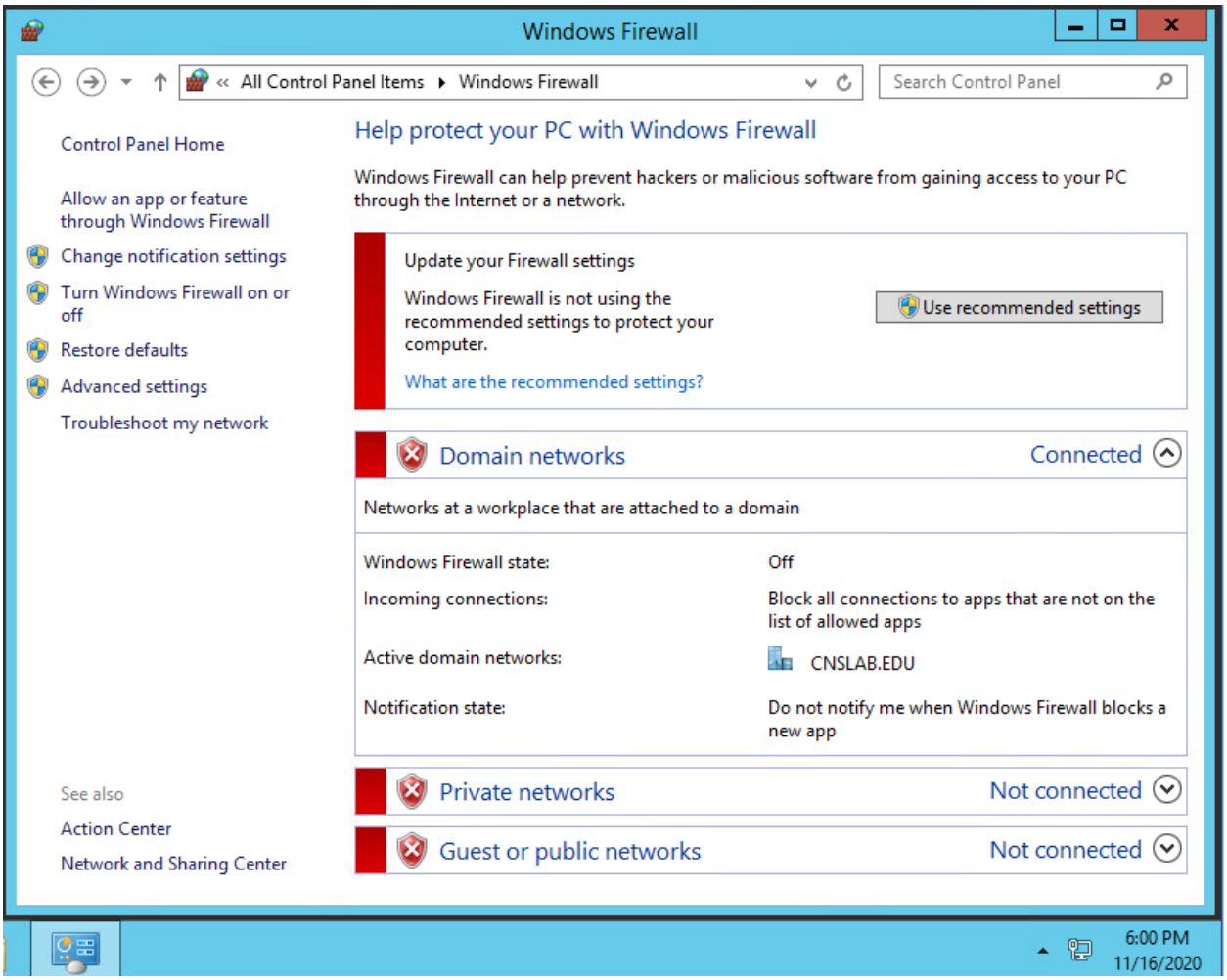


- The **Customize Settings** window appears.
- Set "**Turn off Windows Firewall**" for **Domain**, **Private** and **Public** network settings.
- Move the window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look like



- After taking the screenshot, click on **OK**

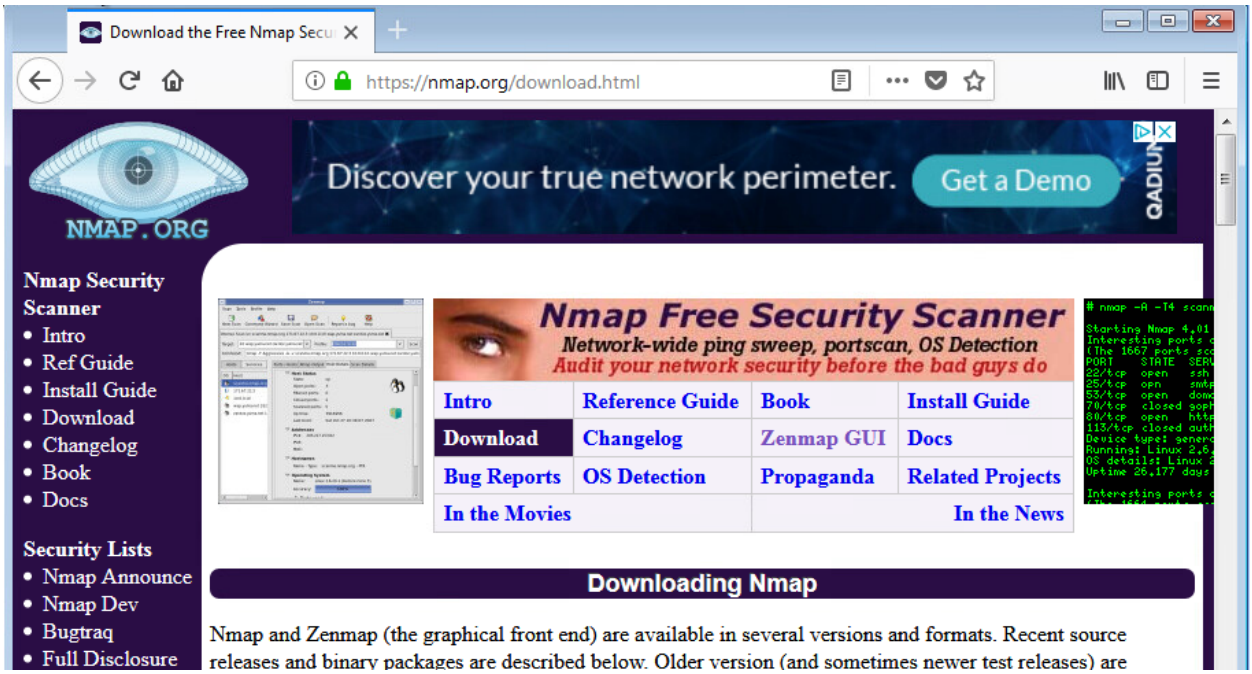
INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***5 POINTS***



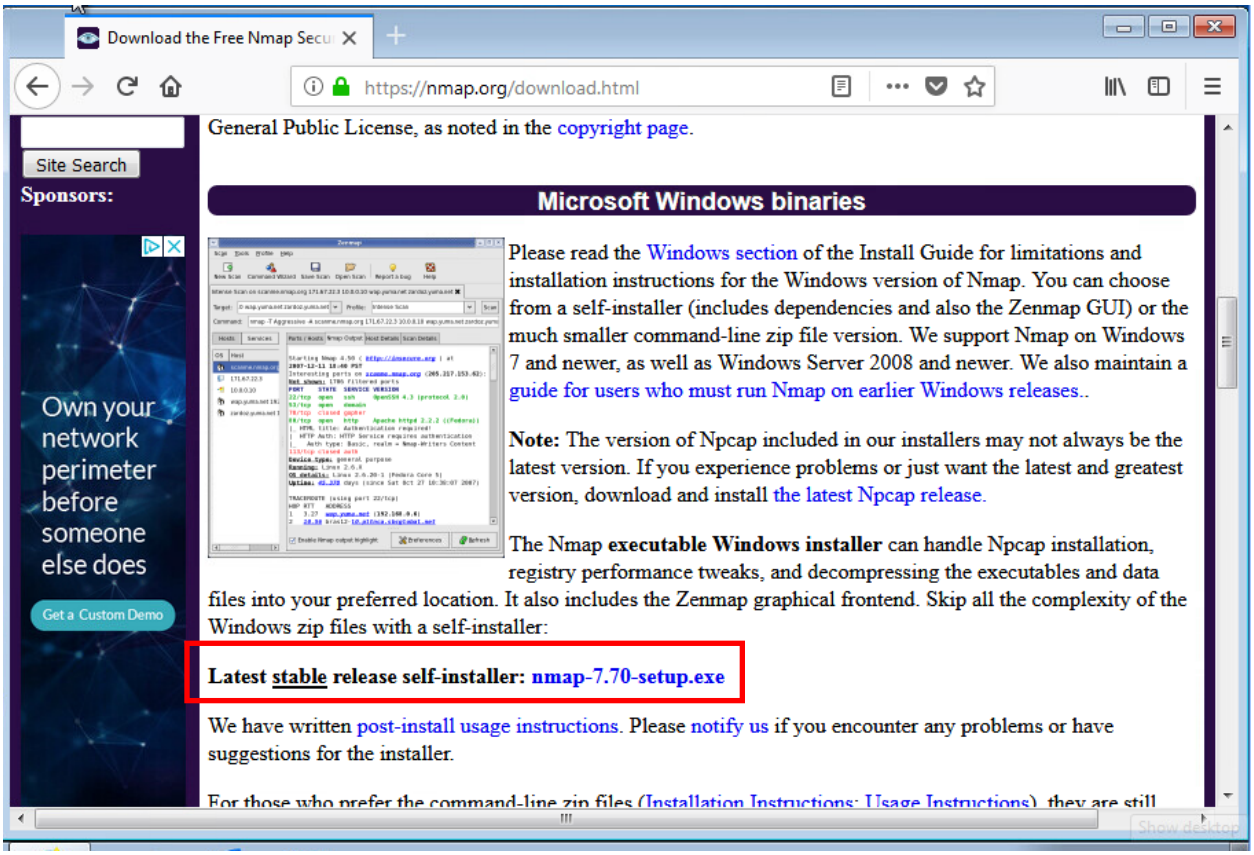
Using zenmap

Now we will install the port scanner utility **zenmap**

- Login to the **CNSLabWin7** VM. The password is "**cnslab**"
- You may need to resize the VM to expose the taskbar at the bottom of the window
- Open **Firefox**
- Go to https://nmap.org/download.html
- The following page appears

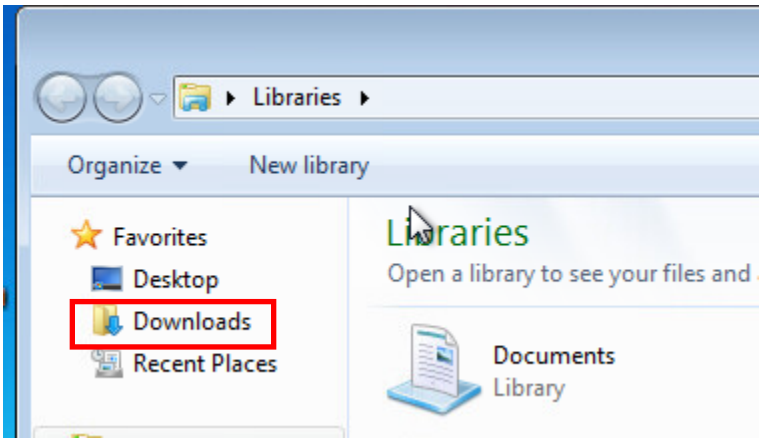

Download the Free Nmap Secu
https://nmap.org/download.html
NMAP.ORG
Discover your true network perimeter.
Get a Demo
Nmap Security Scanner
Intro
Ref Guide
Install Guide
Download
Changelog
Book
Docs
Security Lists
Nmap Announce
Nmap Dev
Bugtraq
Full Disclosure
Nmap Free Security Scanner
Network-wide ping sweep, portscan, OS Detection
Audit your network security before the bad guys do
Intro
Reference Guide
Book
Install Guide
Download
Changelog
Zenmap GUI
Docs
Bug Reports
OS Detection
Propaganda
Related Projects
In the Movies
In the News
Downloading Nmap
Nmap and Zenmap (the graphical front end) are available in several versions and formats. Recent source releases and binary packages are described below. Older version (and sometimes newer test releases) are

- Scroll down until you see the "**Microsoft Windows binaries"** section
- Click on "**Latest stable release self-installer**"

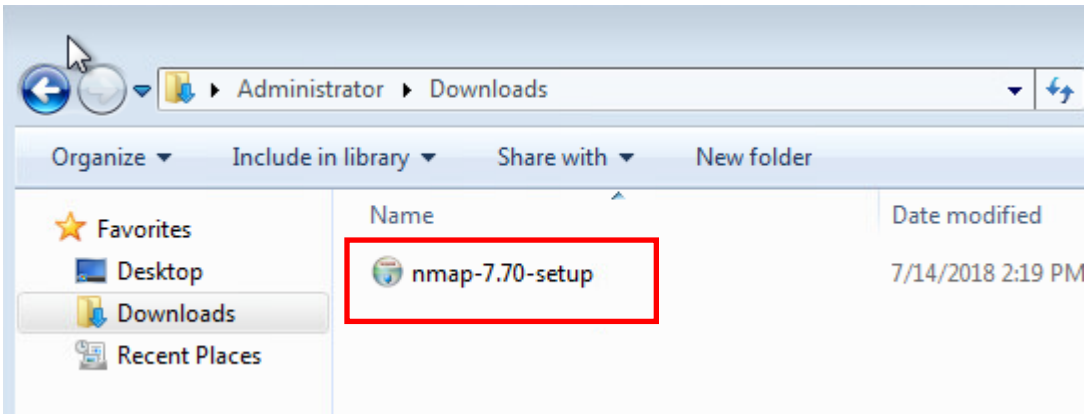


- The following pop-up appears. NOTE: The displayed version number may be different
- Click on **Save File**

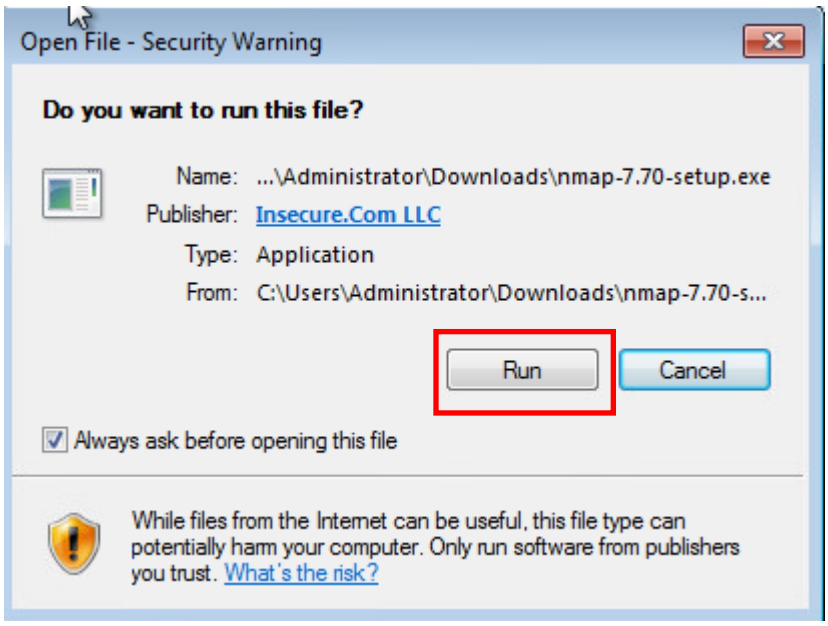
- Close **Firefox**
- Open **File Explorer**
- Click **Downloads**



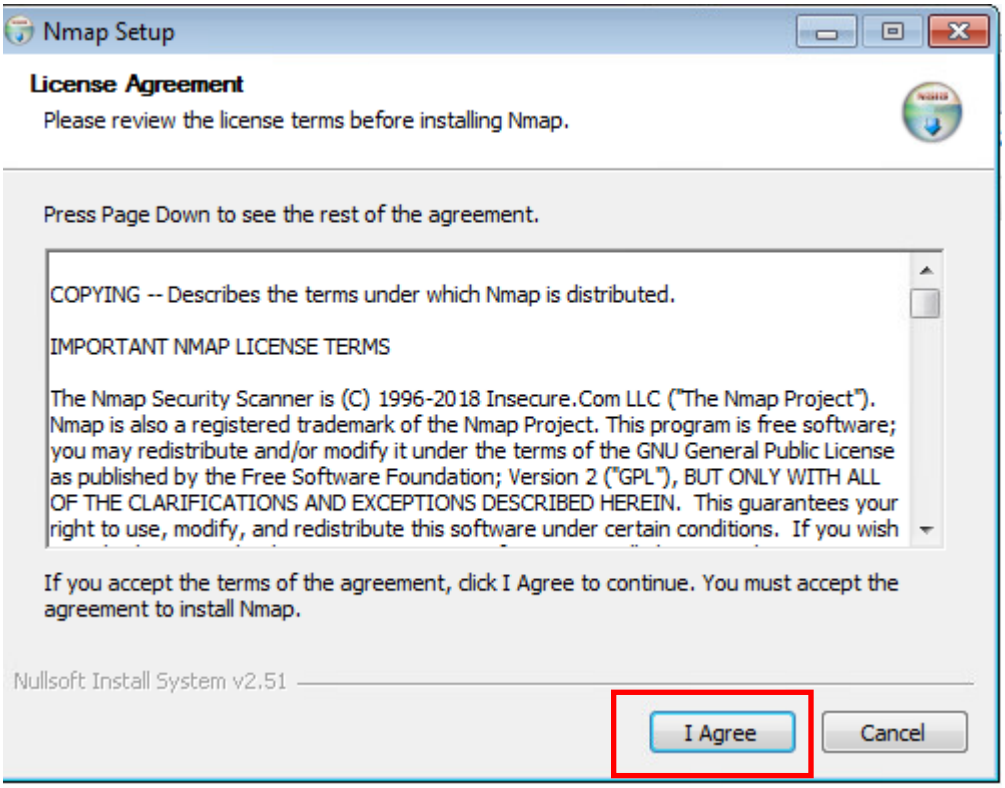
- The **Downloads** folder window appears
- Double-click on **nmap-7.70-setup** to install the software. NOTE: The filename might show a different version number



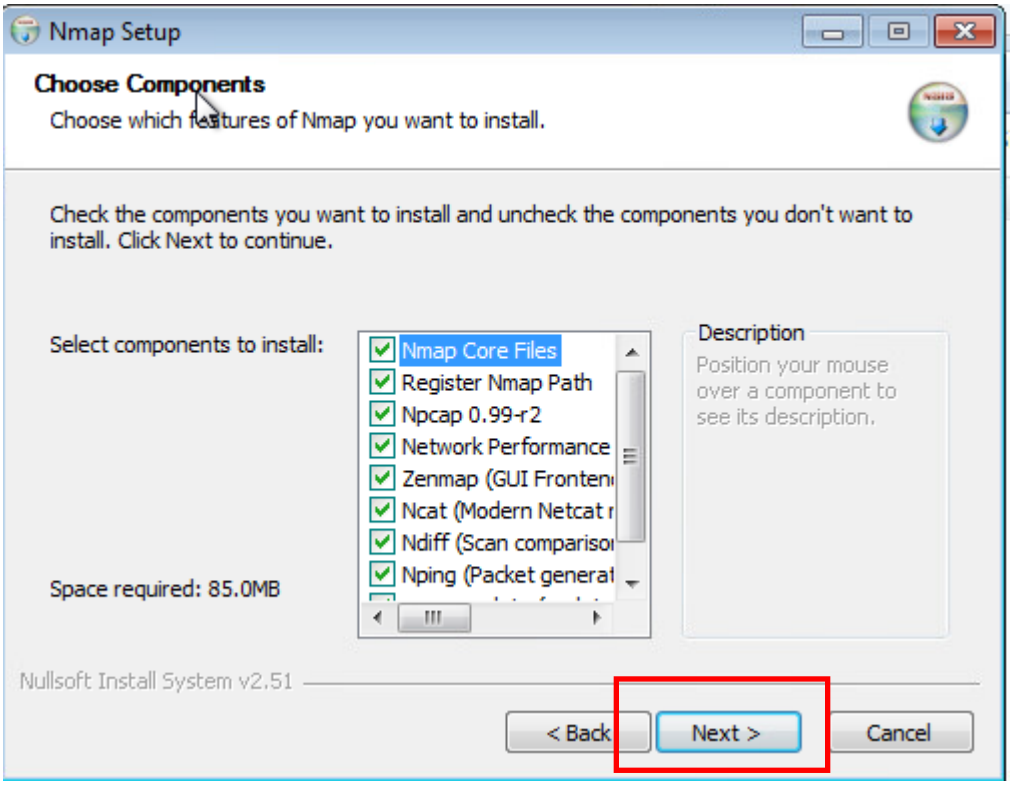
- A **Security Warning** window appears
- Click **Run**



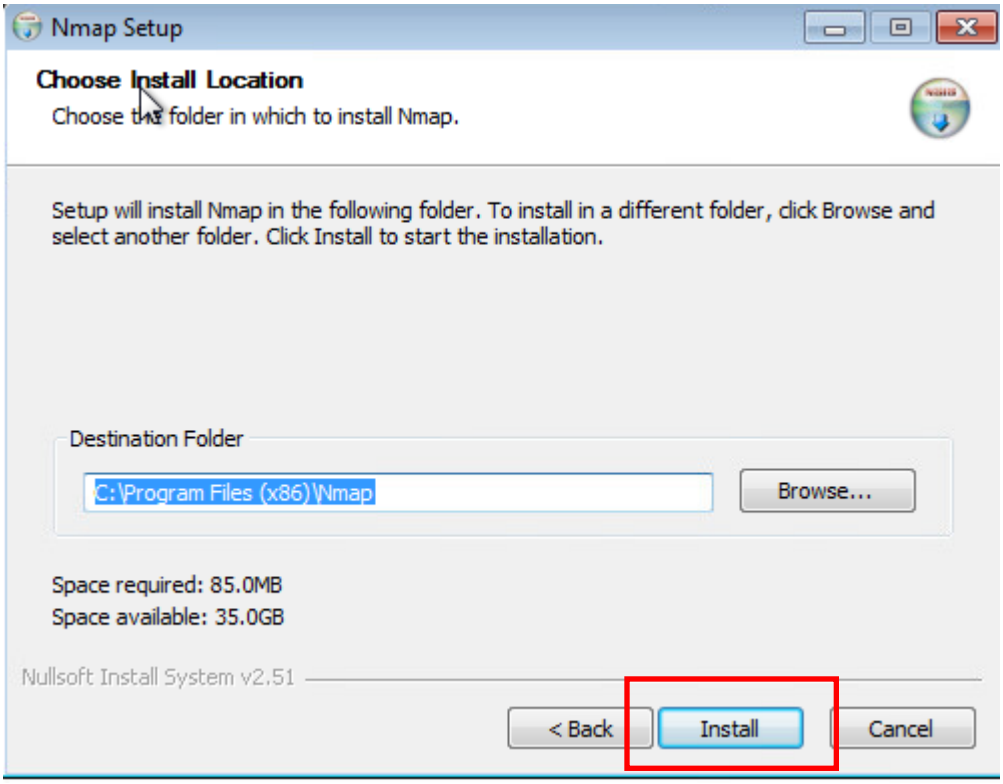
- The :**Nmap "License Agreement"** window for **Nmap** appears
- Click on **I Agree**



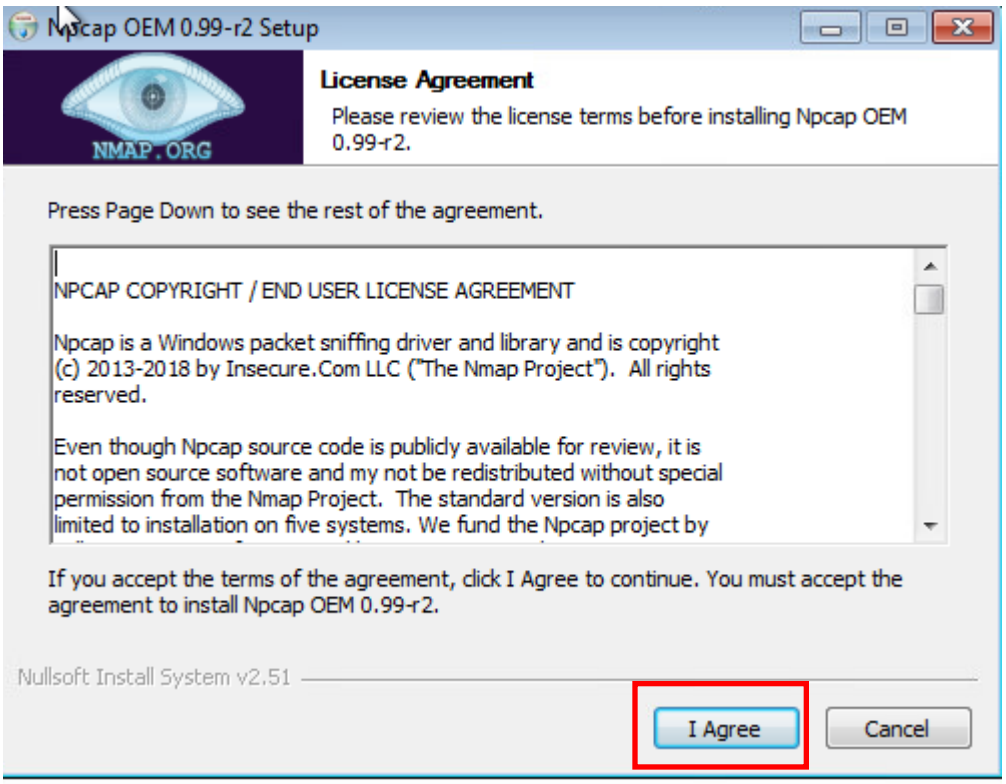
- The "**Chose Components**" window for **Nmap** appears
- Click on **Next**



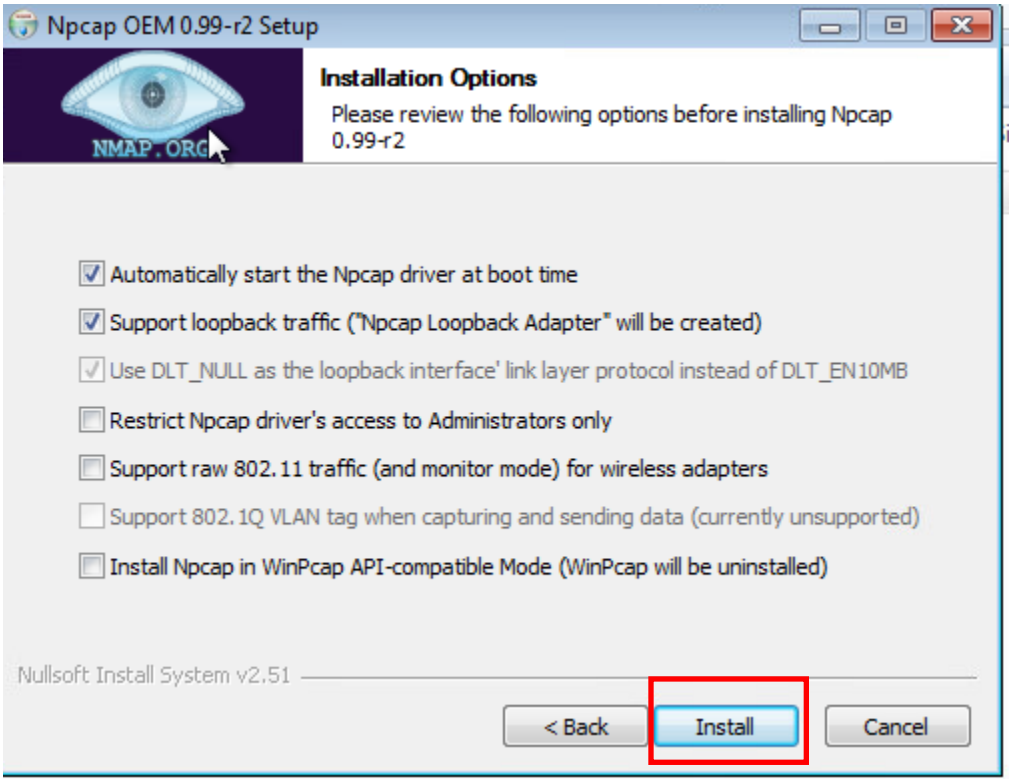
- The "**Choose Install Location**" window for **Nmap** appears
- Click on **Install**



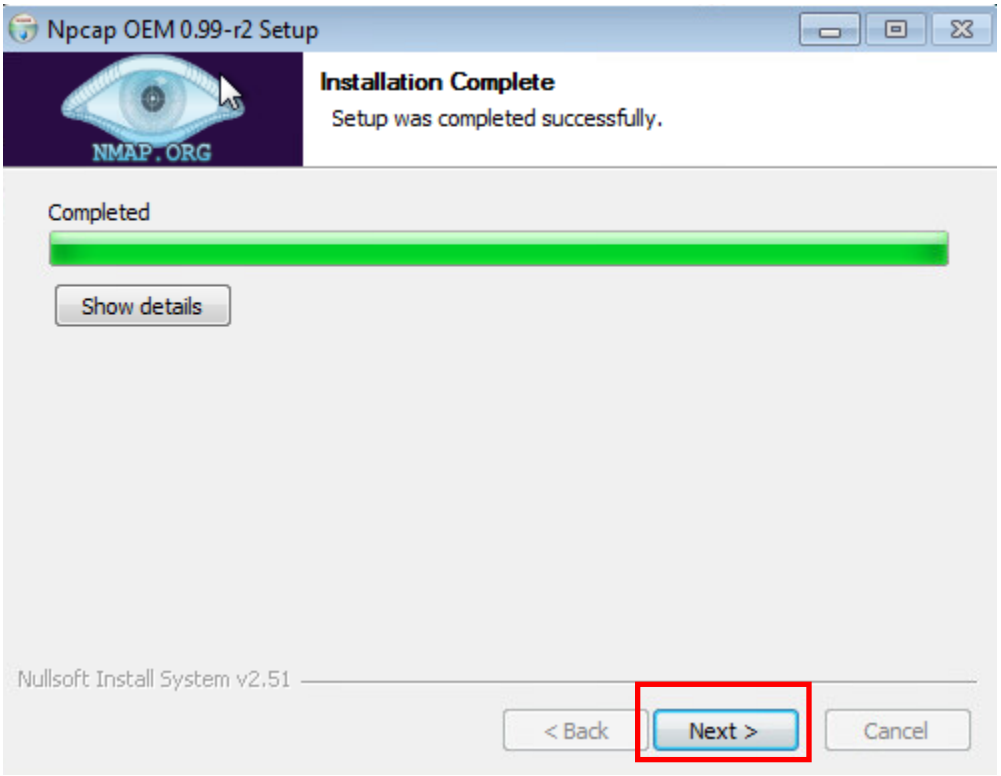
- After the **Nmap** is installed , the **License Agreement** window for **Npcap** appears
- Click **I Agree**



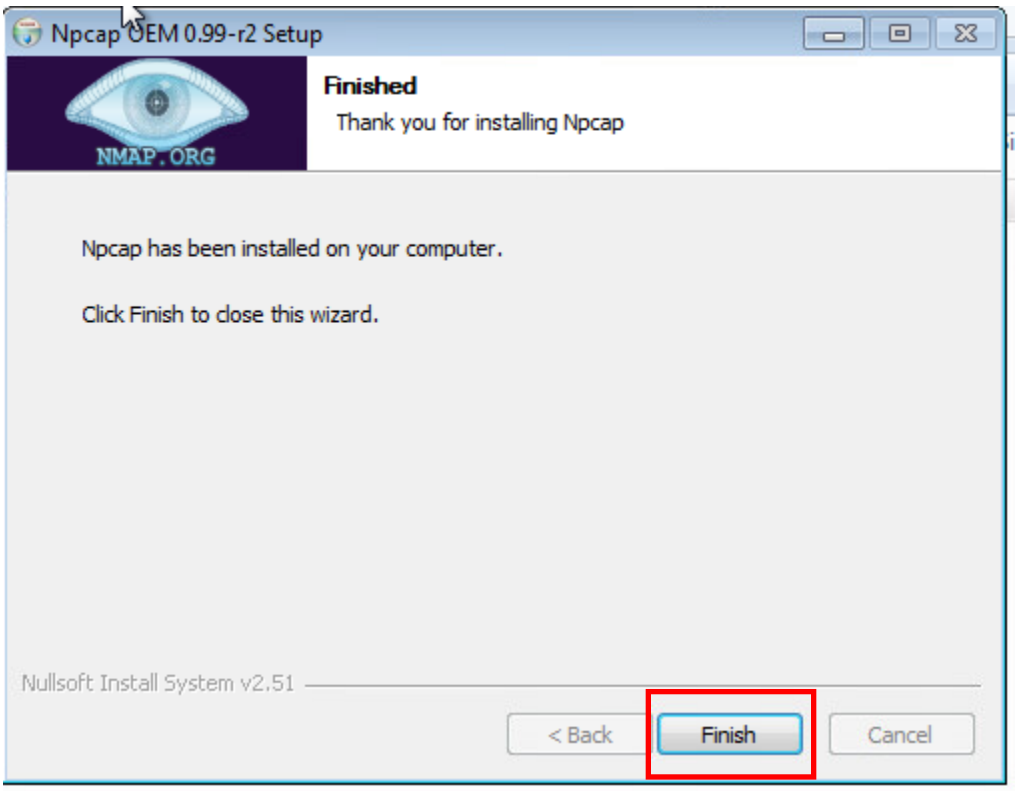
- The **Installation Options** for **Npcap** window appears
- Click on **Install**



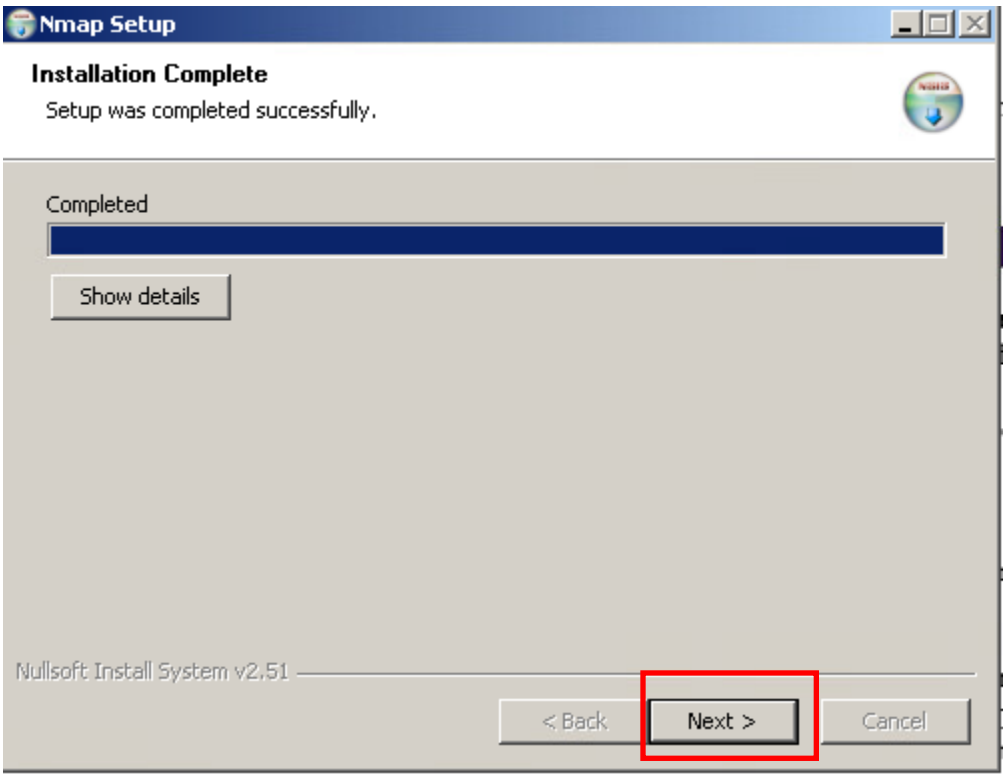
- The **Installation Complete** window for **Npcap** appears
- Click on **Next**



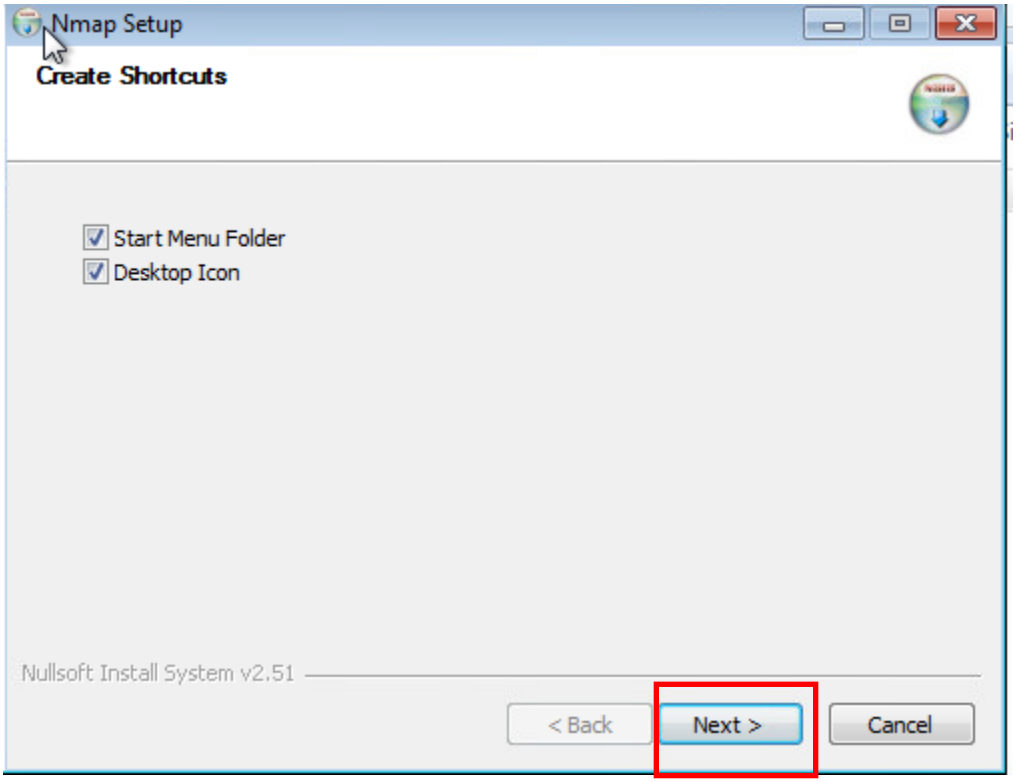
- The **Finished** window for **Npcap** appears
- Click on **Finish**



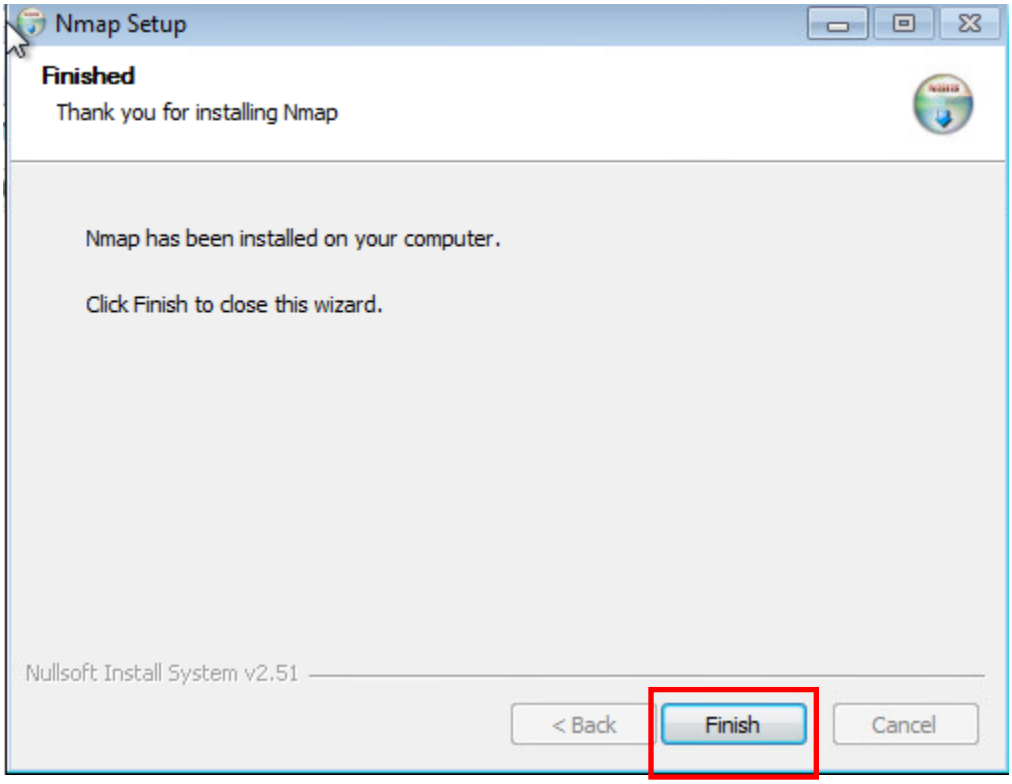
- The **Installation Complete** window for **Nmap** appears
- Click on **Next**



- The **Create Shortcuts** window for **Nmap** appears
- Click on **Next**



- The **Finished** window for **Nmap** appears
- Click on **Finish**

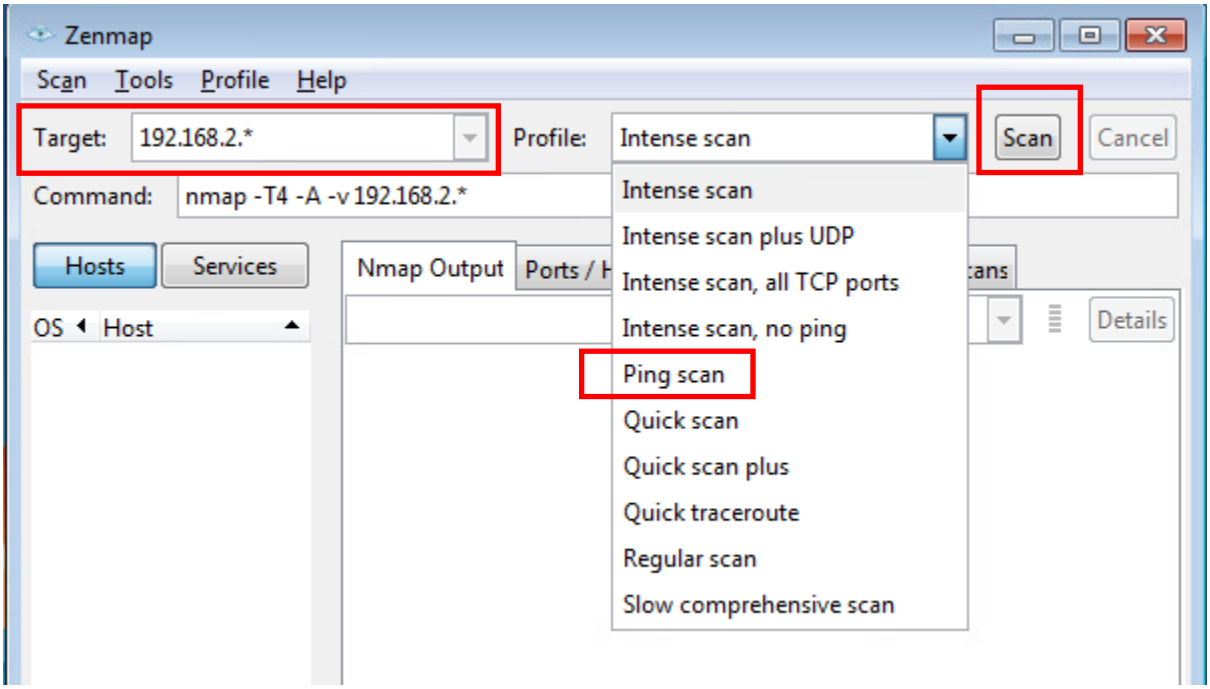


- Close **File Explorer**

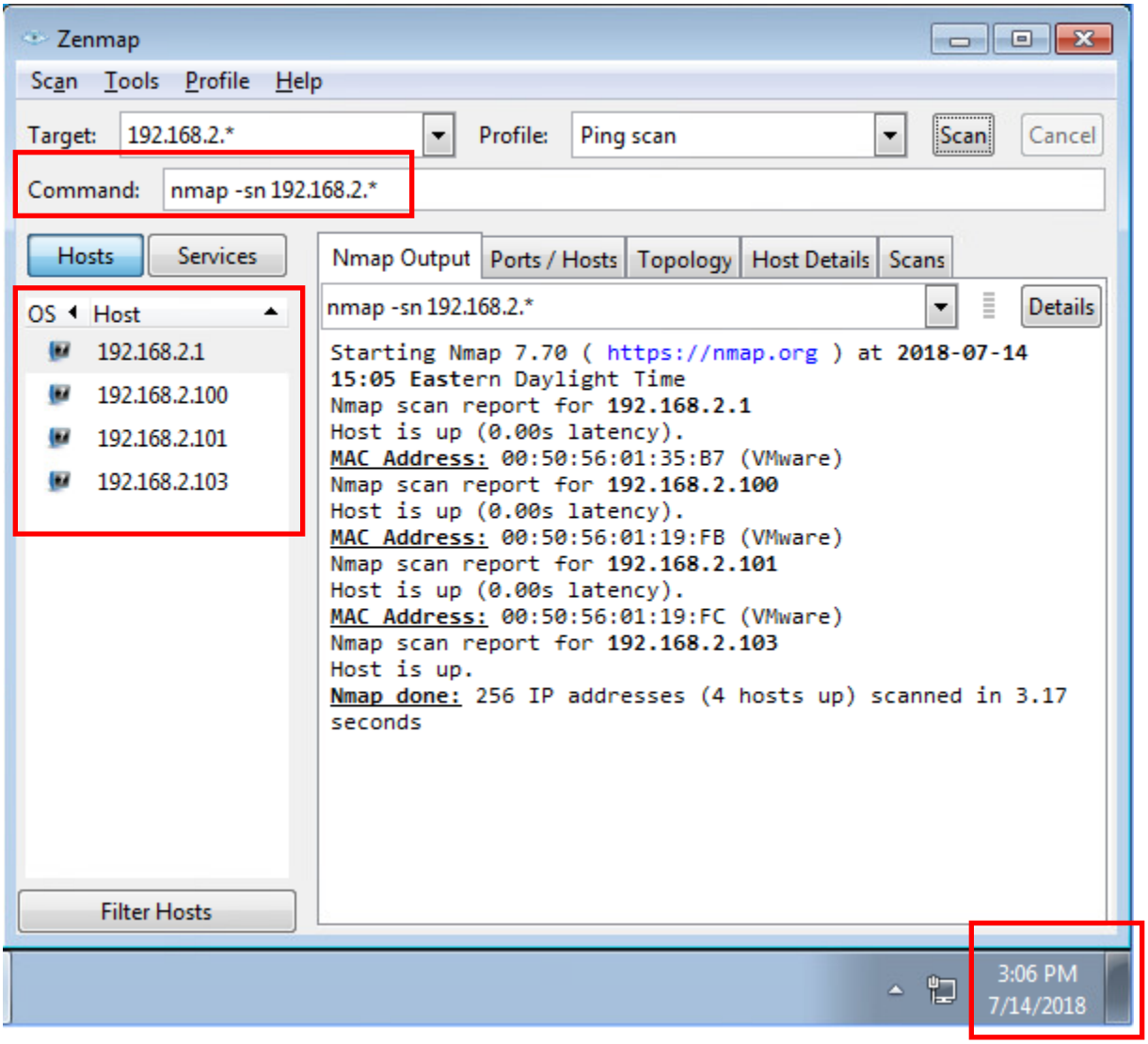
Now we can use **zenmap** to probe the systems on the vApp network

- Double-click the **Nmap-Zenmap-GUI** icon on the Desktop

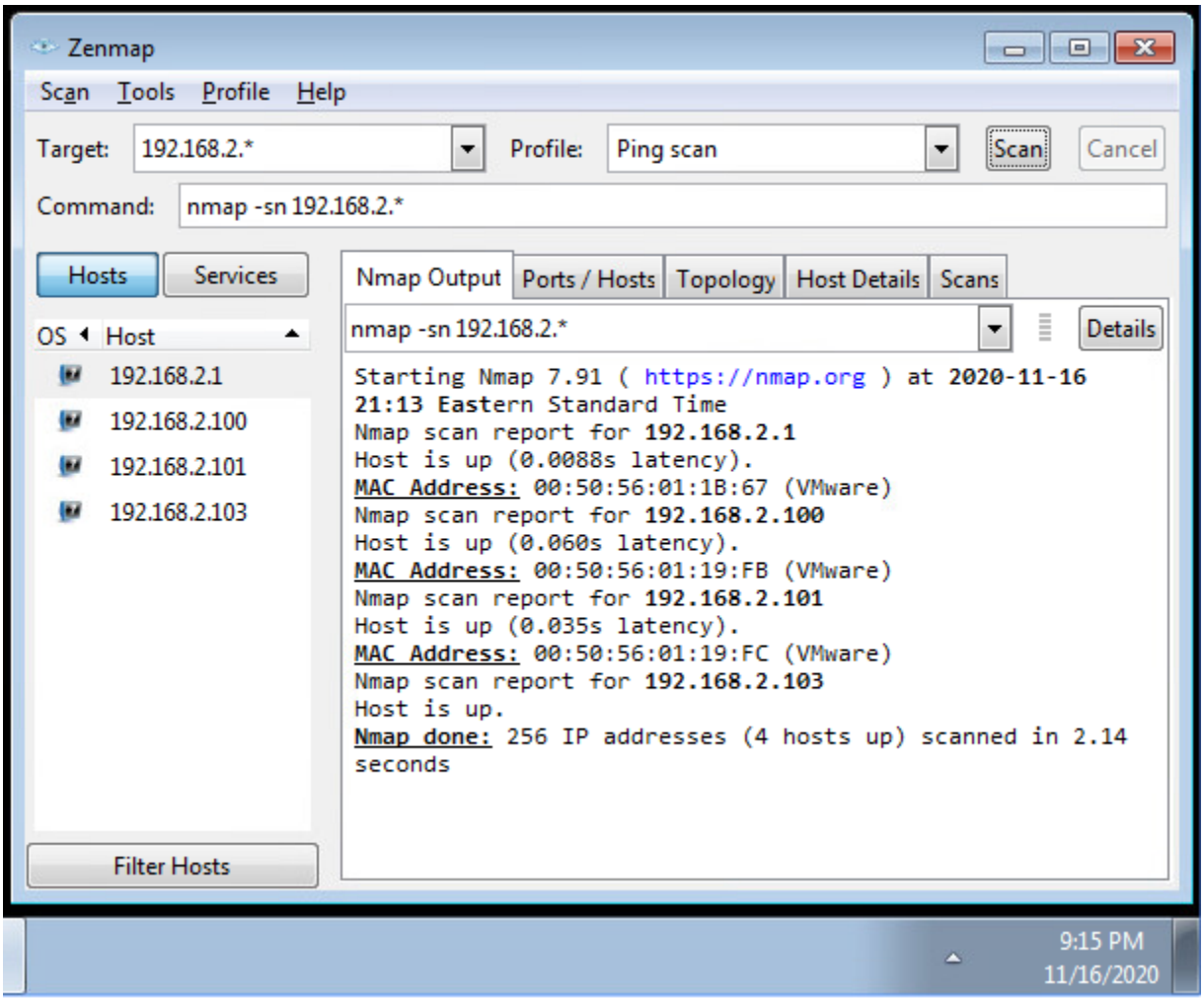
- The **Zenmap** window appears
- Enter **192.168.2.*** in the Target field. NOTE: The last character is an asterisk. We want **Zenmap** to scan the local subnet.
- Click on the arrow to the right of the **Profile** field to show the drop-down list
- Select **ping scan**.
- Click on **Scan**



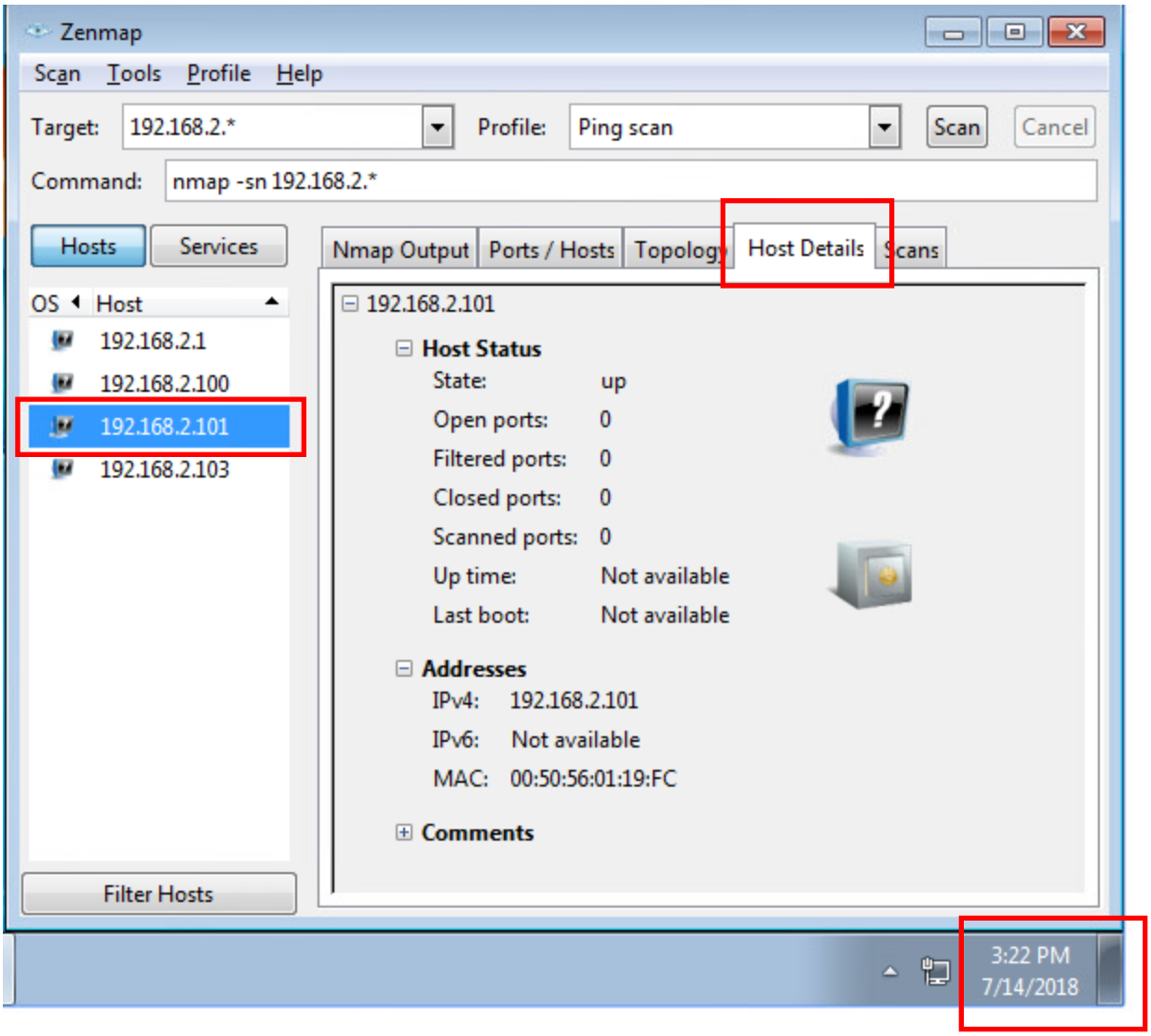
- The left pane shows four hosts. The icons to the left of the IP addresses indicate their probe status. Since only a ping scan was performed, we don't know anything but their IP addresses for now
- Make the window shorter to make screenshots fit better in the document
- NOTE: The value in the **Command** field has changed. It is now the **nmap** command you run to perform this type of scan; i.e. **nmap –sn 192.168.2.***
- Move the window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look like



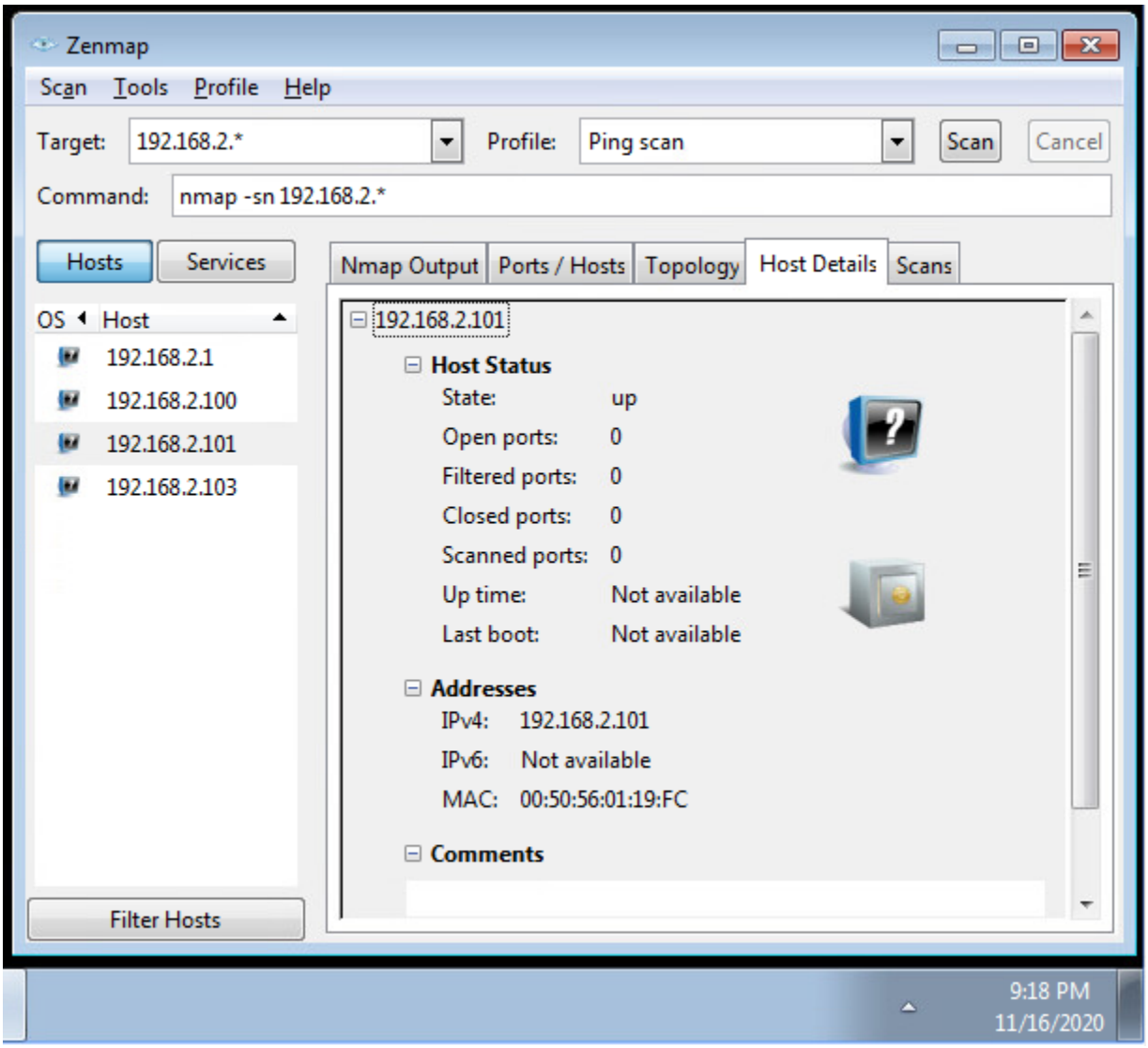
INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***5 POINTS***



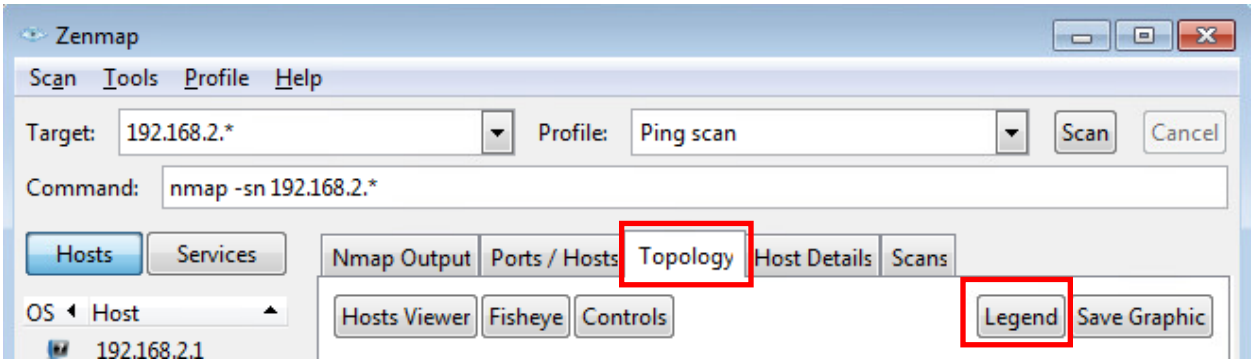
- Select **192.168.2.101** in the left pane, to select the **CNSLabSvrMember** system
- Click the **Host Details** folder tab
- The right pane is updated to show host information about **CNSLabSvrMember**. NOTE: If needed, make the window taller to show **Host Status**, **Addresses** and **Comments**
- Move the window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look like


Zenmap
Scan Tools Profile Help
Target: 192.168.2.*
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 192.168.2.*
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
OS
Host
192.168.2.1
192.168.2.100
192.168.2.101
192.168.2.103
192.168.2.101
Host Status
State: up
Open ports: 0
Filtered ports: 0
Closed ports: 0
Scanned ports: 0
Up time: Not available
Last boot: Not available
Addresses
IPv4: 192.168.2.101
IPv6: Not available
MAC: 00:50:56:01:19:FC
Comments
Filter Hosts
3:22 PM
7/14/2018

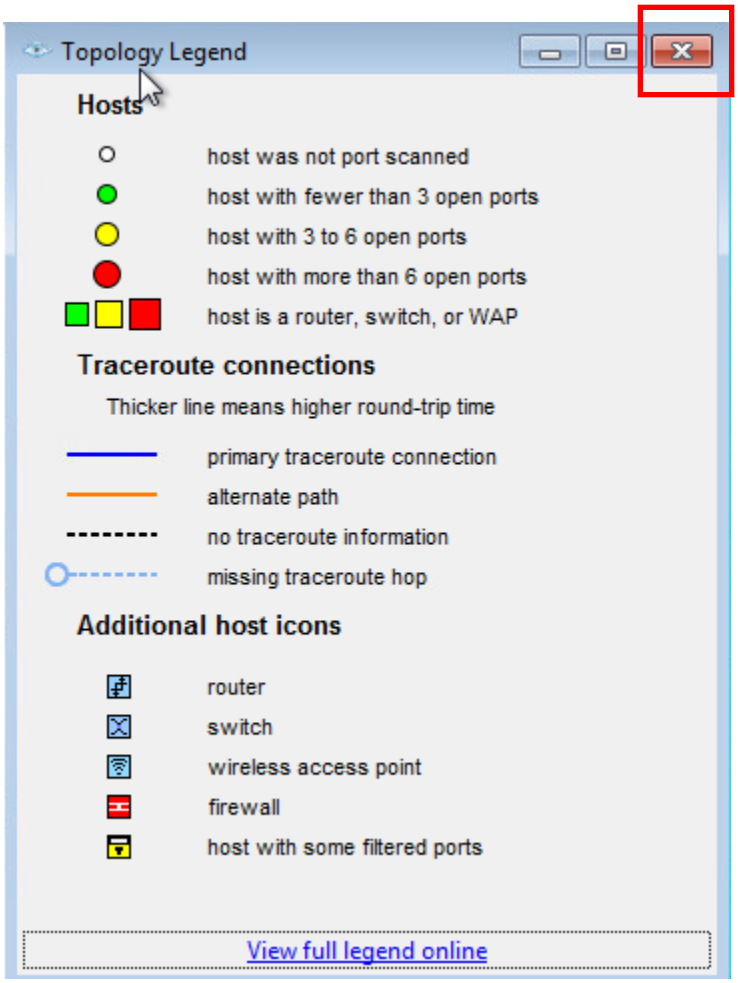
INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***5 POINTS***



- Click on "**Topology**" folder tab
- The right pane changes to show a diagram of the systems on the network
- Click on **Legend**

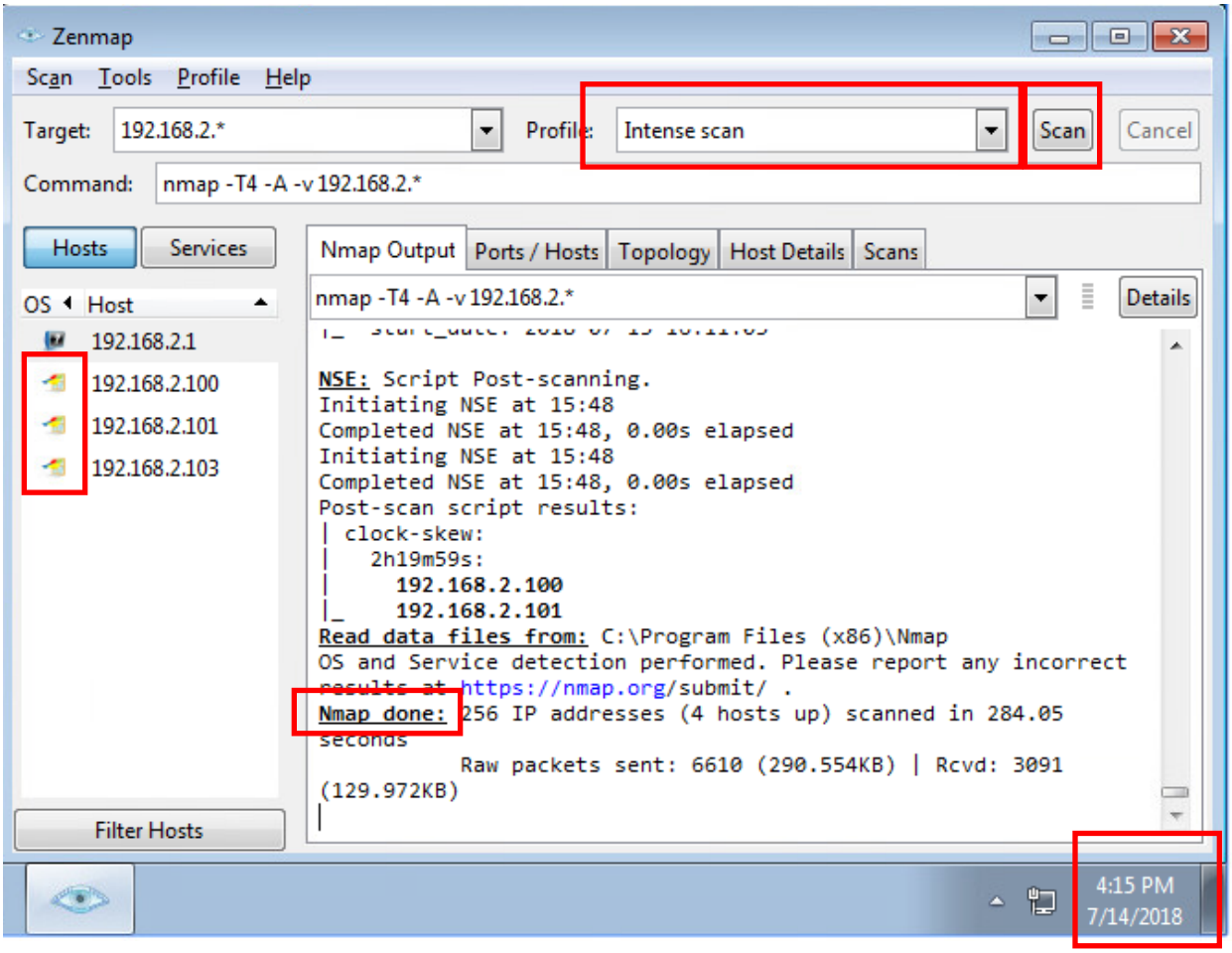


- The **Topogy Legend** appears

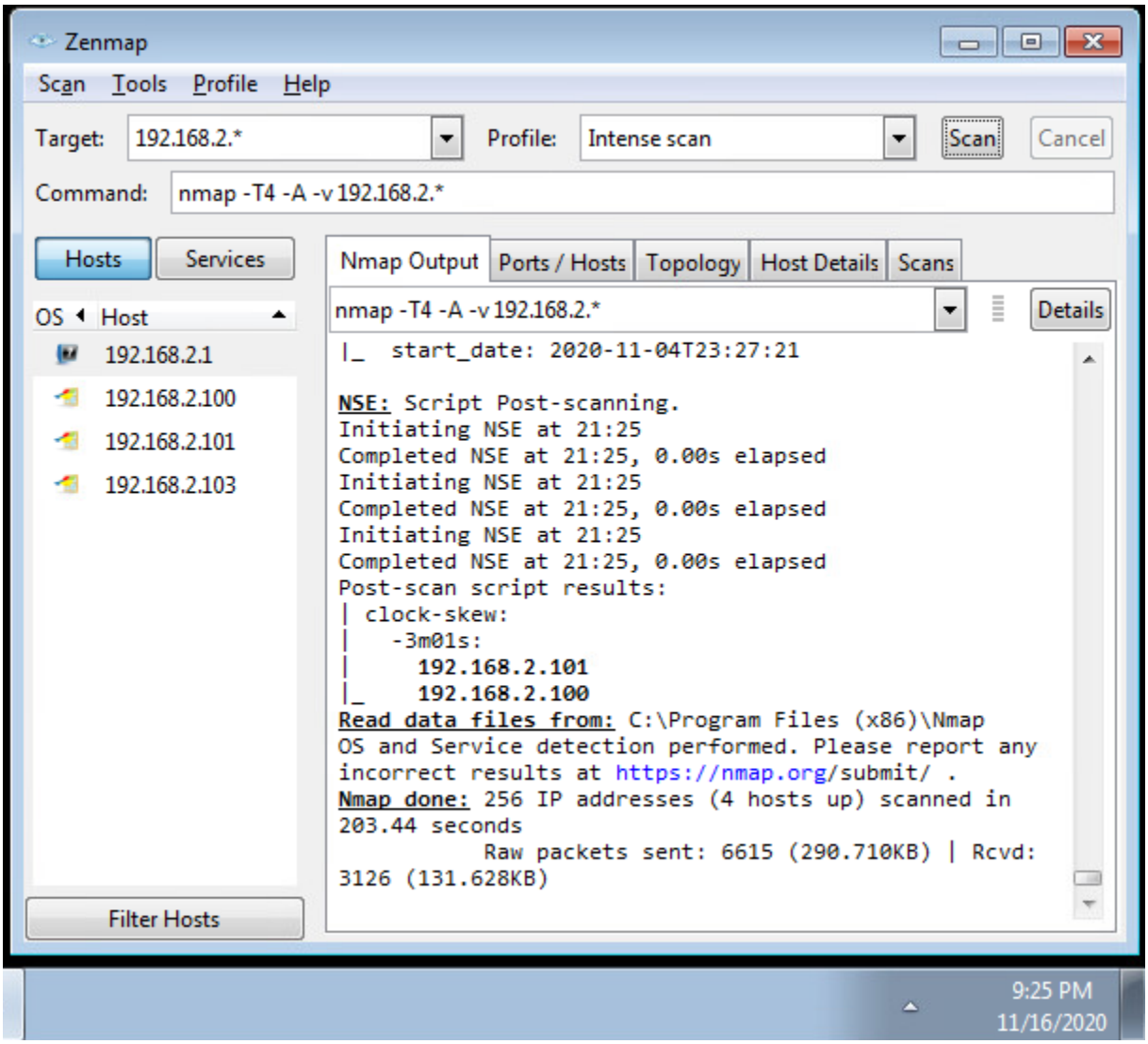


- Click on "**X**" to close

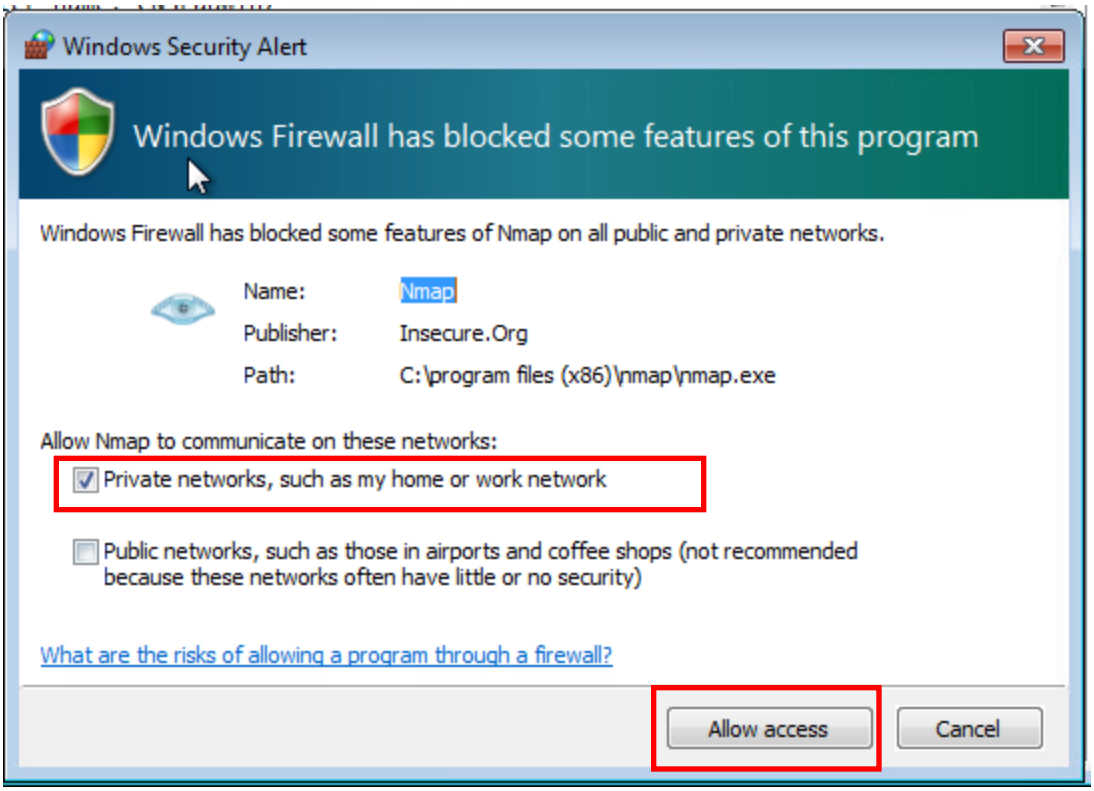
- Change the value in the **Profile** field to **Intense Scan**
- NOTE: The value in the **Command** field has changed. It is now the **nmap** command you run to perform this type of scan; i.e. **nmap –T4 –A –v 192.168.2.***
- Click on **Scan**.
- NOTE: The **Intense Scan** takes about 5 minutes
- When you see "**Nnap done: ….**" in the right pane, then the scan has completed
- NOTE: The icons in the left pane have changed for **192.168.2.100**, **192.168.2.101** and **192.168.2.103** but not for **192.168.2.1**.
- Move the window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look like



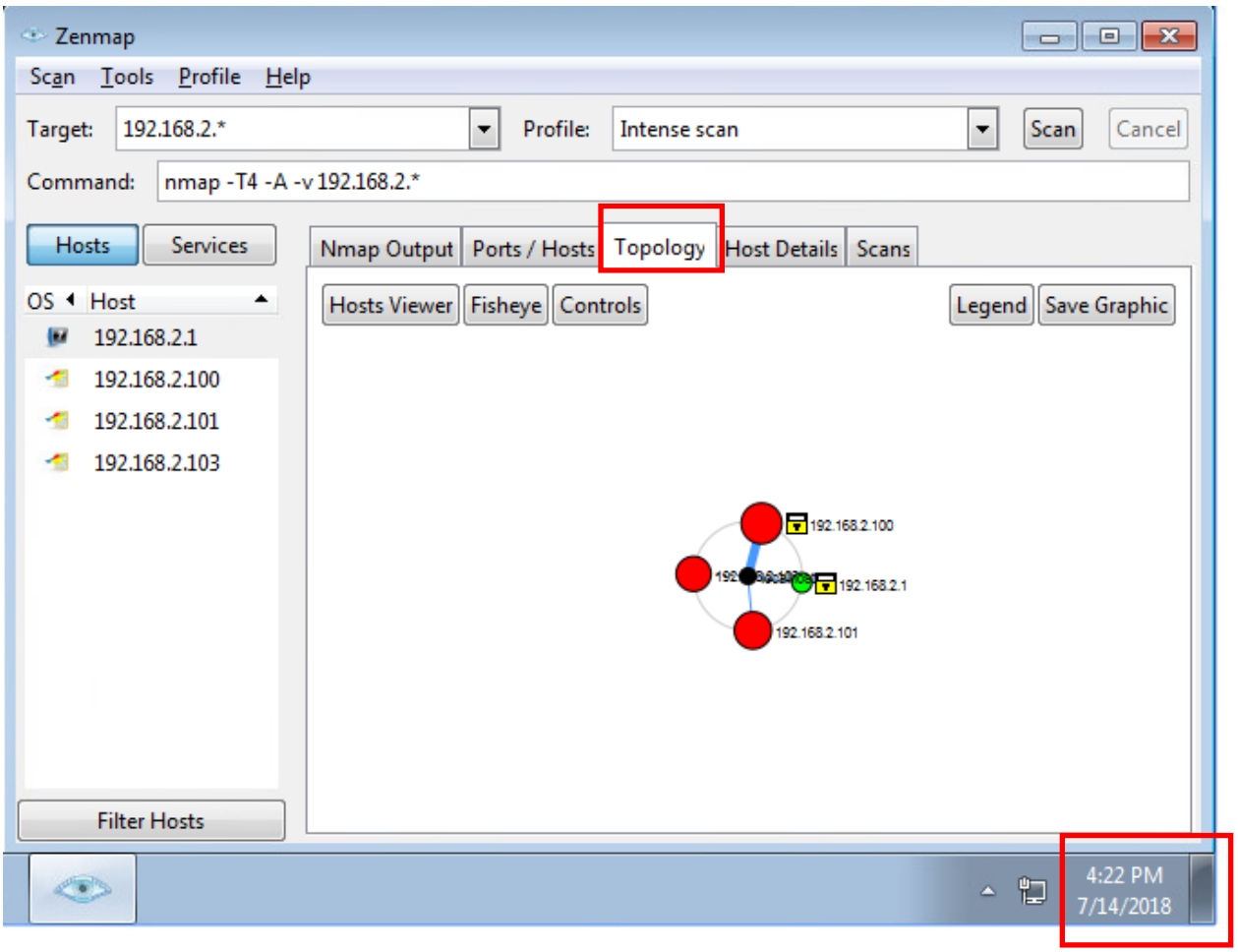
INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***5 POINTS***



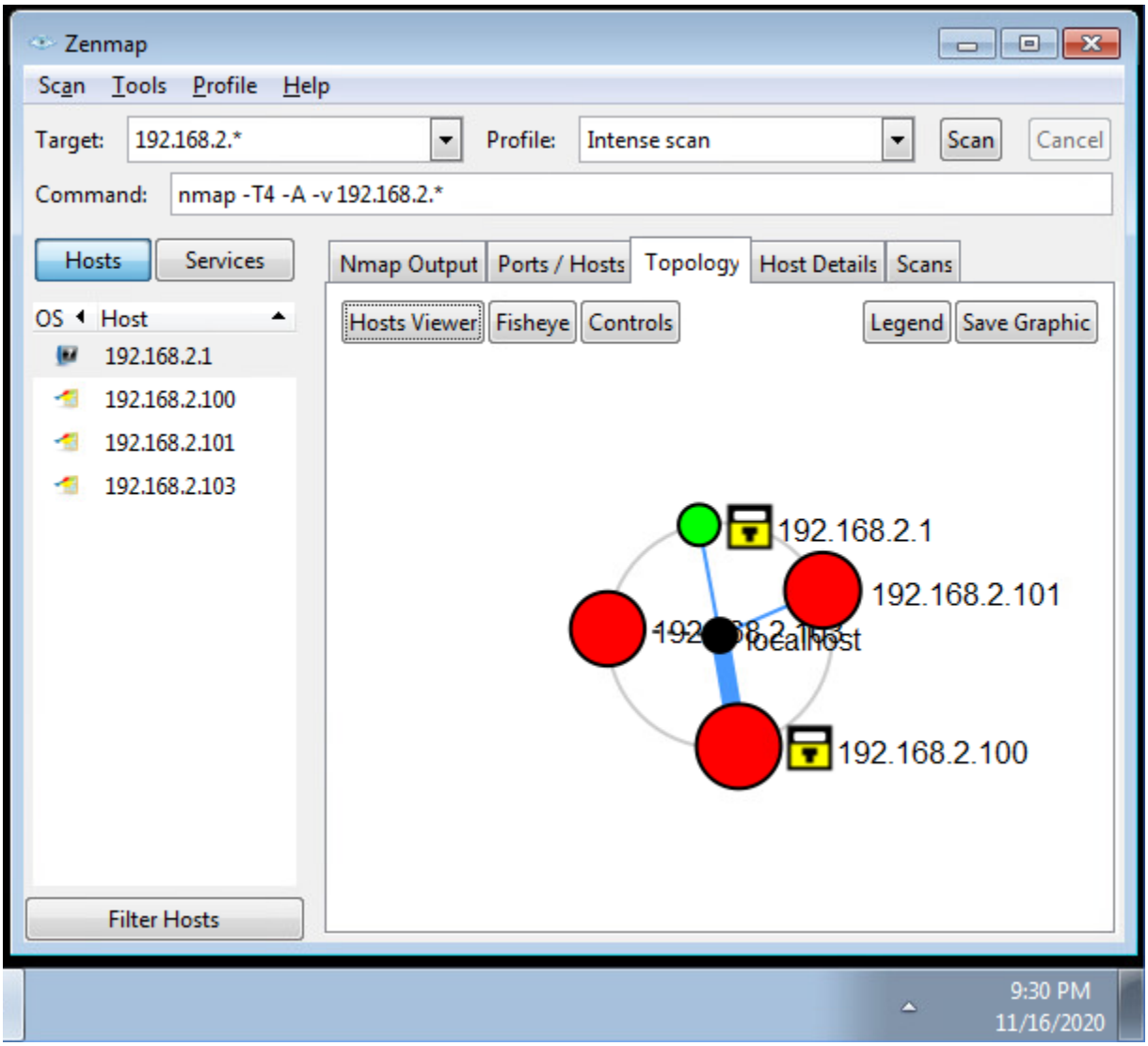
- If you see **Windows Security Alert** screen, click on **Allow Access**


Windows Security Alert
Windows Firewall has blocked some features of this program
Windows Firewall has blocked some features of Nmap on all public and private networks.
Name: Nmap
Publisher: Insecure.Org
Path: C:\program files (x86)\nmap\nmap.exe
Allow Nmap to communicate on these networks:
Private networks, such as my home or work network
Public networks, such as those in airports and coffee shops (not recommended because these networks often have little or no security)
What are the risks of allowing a program through a firewall?
Allow access
Cancel

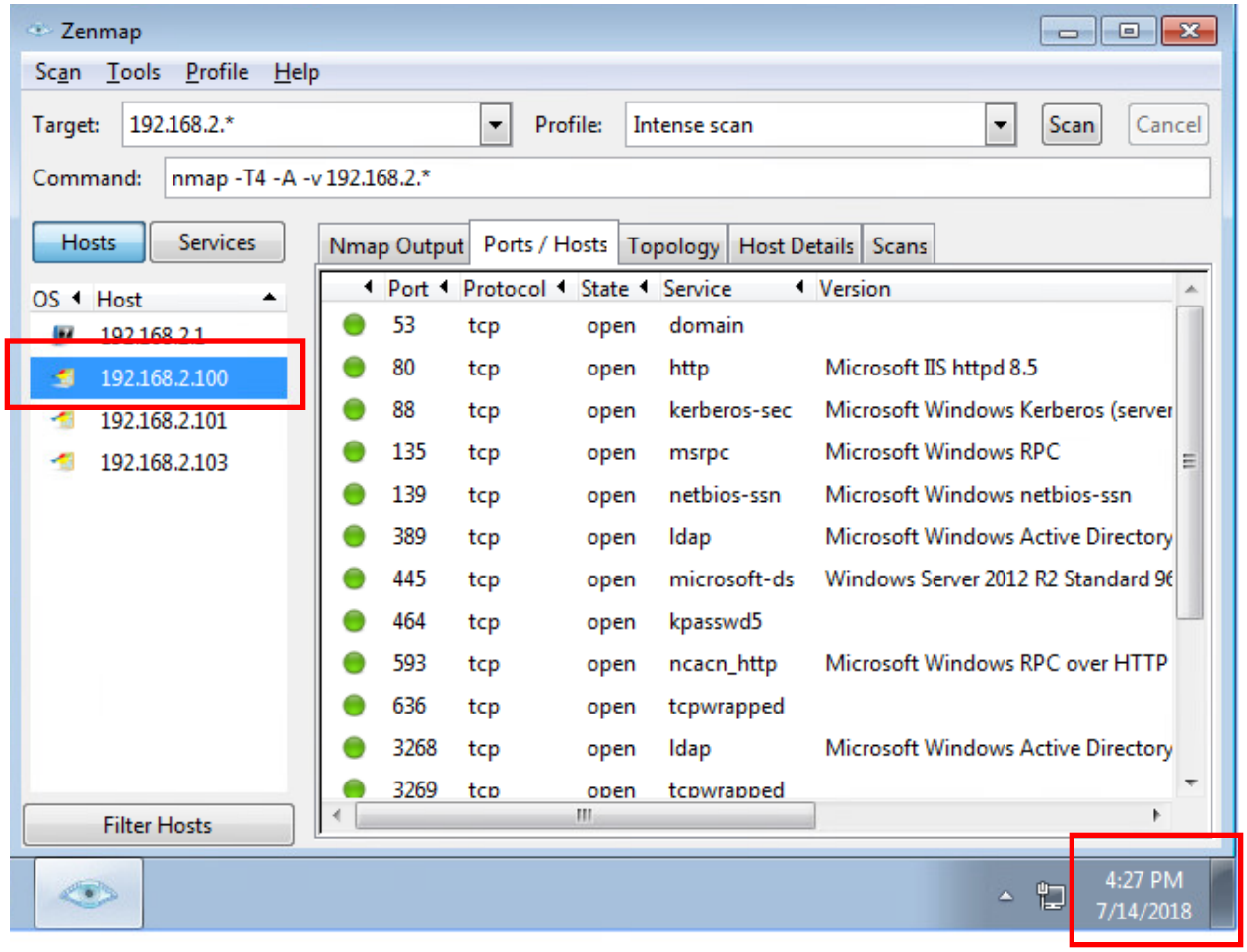
- Click on "**Topology**" folder tab
- The right pane changes to show a diagram of the systems on the network
- NOTE: The red circles indicate systems that had more than 6 open ports. The green circle indicates the system (**192.168.2.1)** that fewer than 3 open ports
- Move the window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look something like



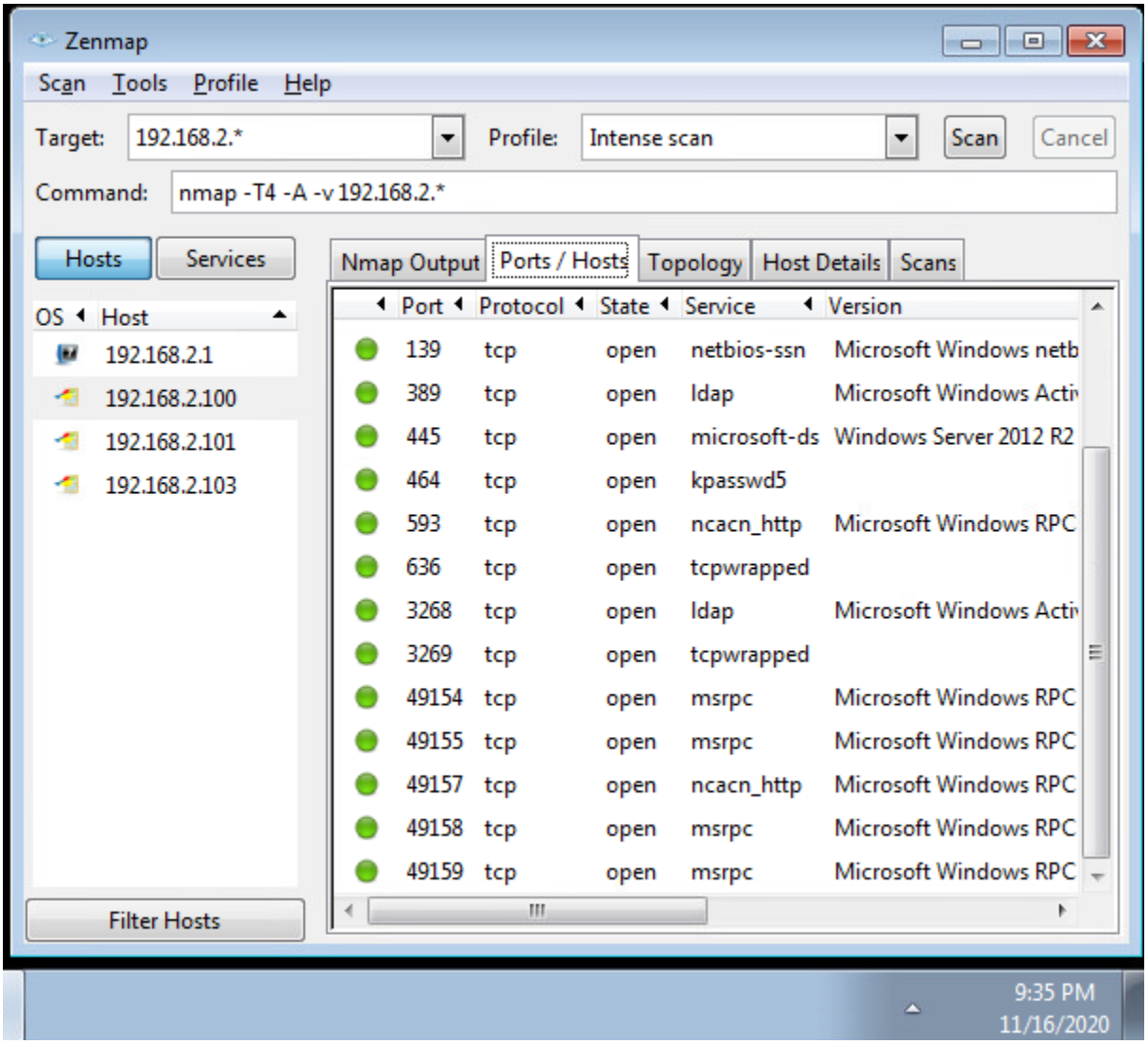
INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***10 POINTS***



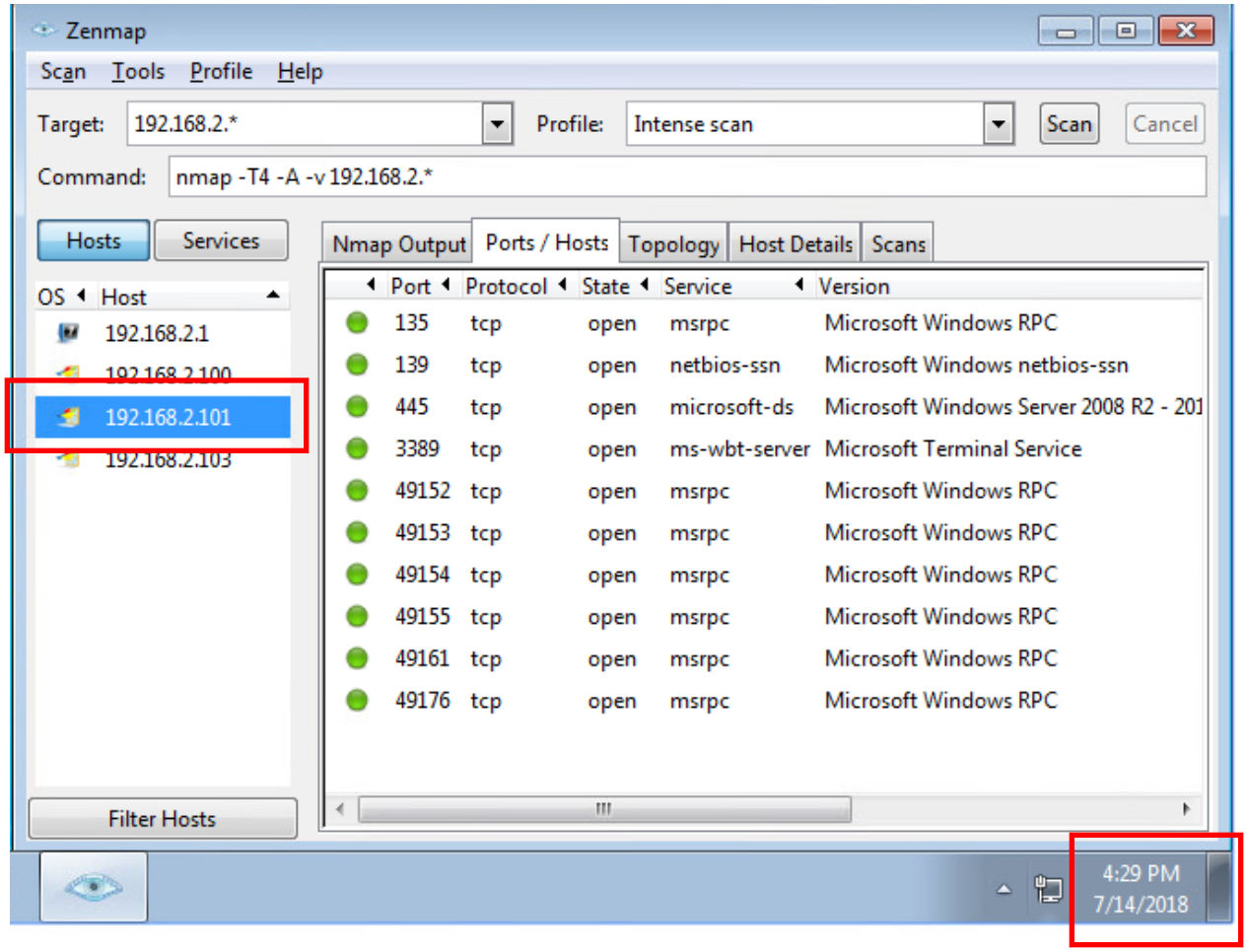
- Click on **192.168.1.100** in the left pane, to select the **CNSLabSvrDC** system
- Click on " **Port / Hosts** " folder tab
- The right pane changes to show port information about **CNSLabSvrDC**
- Move the window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look like


Zenmap
Scan Tools Profile Help
Target: 192.168.2.*
Profile: Intense scan
Scan
Cancel
Command: nmap -T4 -A -v 192.168.2.*
Hosts
Services
OS Host
192.168.2.1
192.168.2.100
192.168.2.101
192.168.2.103
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Port Protocol State Service Version
53 tcp open domain
80 tcp open http Microsoft IIS httpd 8.5
88 tcp open kerberos-sec Microsoft Windows Kerberos (server
135 tcp open msrpc Microsoft Windows RPC
139 tcp open netbios-ssn Microsoft Windows netbios-ssn
389 tcp open ldap Microsoft Windows Active Directory
445 tcp open microsoft-ds Windows Server 2012 R2 Standard 96
464 tcp open kpasswd5
593 tcp open ncacn_http Microsoft Windows RPC over HTTP
636 tcp open tcpwrapped
3268 tcp open ldap Microsoft Windows Active Directory
3269 tcp open tcpwrapped
4:27 PM
7/14/2018

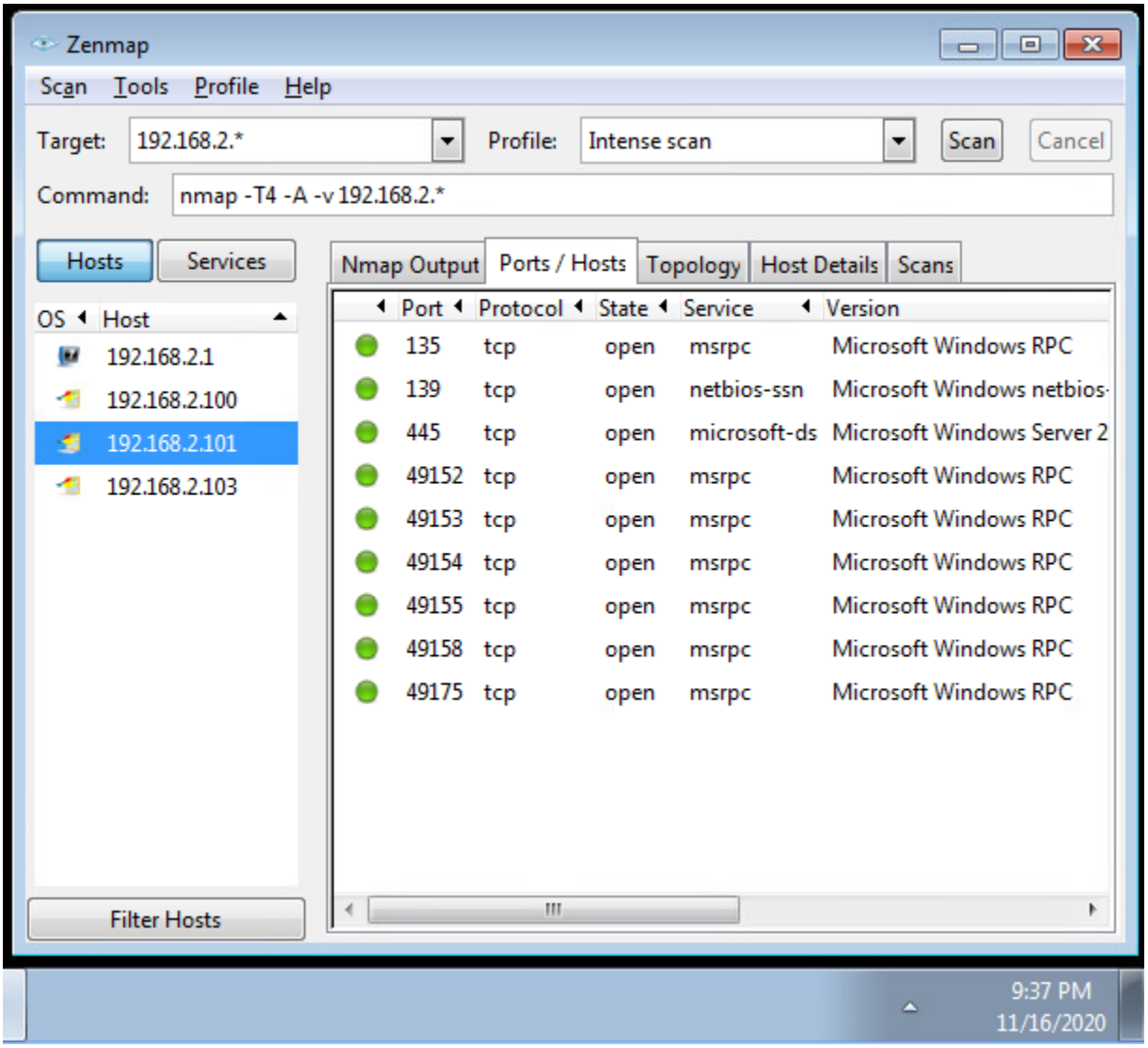
INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***10 POINTS***



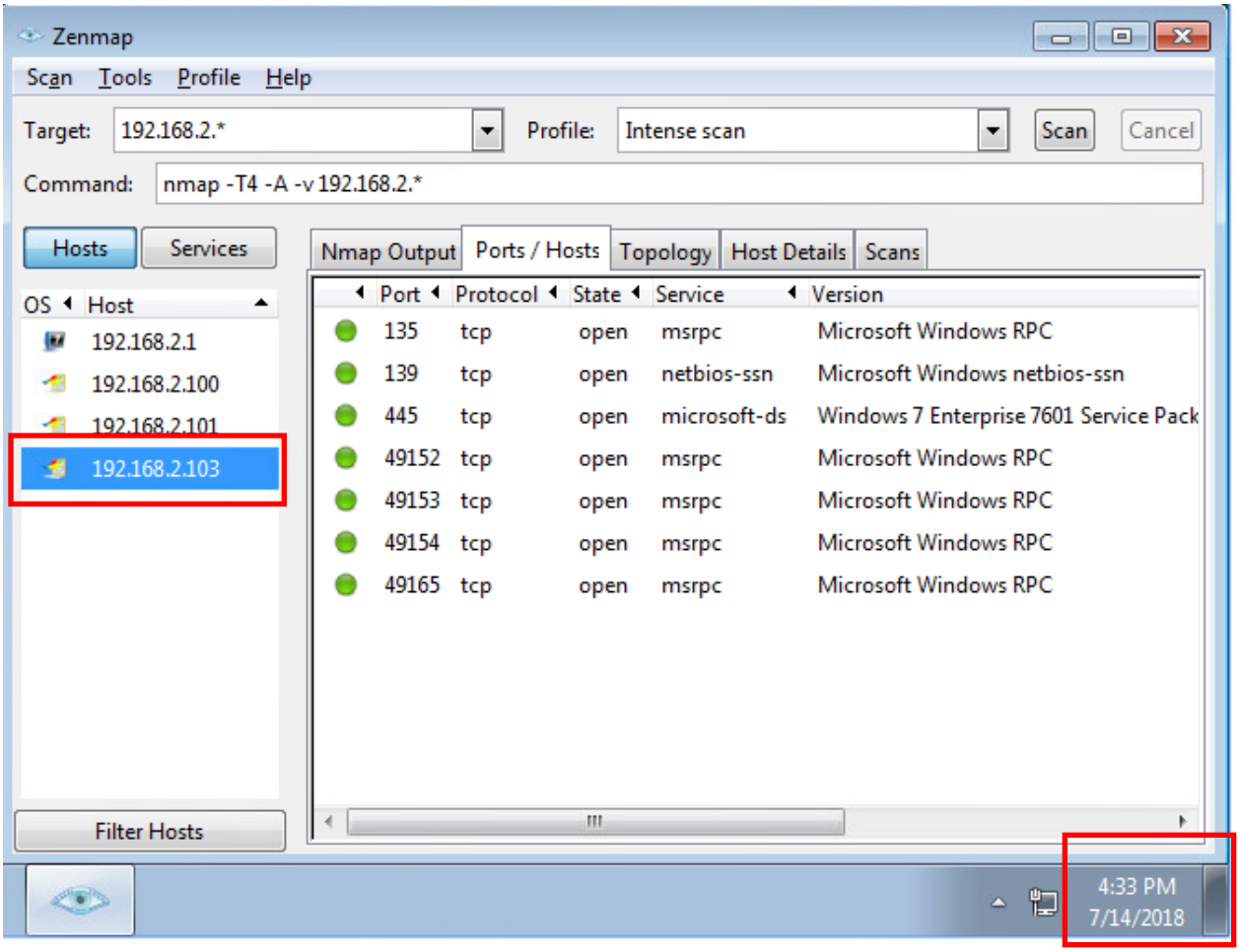
- Click on **192.168.1.101** in the left pane, the **CNSLabSvrMember** system
- Click on "**Port / Hosts**" folder tab
- The right pane changes to show port information about **CNSLabSvrMember**
- Move the window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look like


Zenmap
Scan Tools Profile Help
Target: 192.168.2.*
Profile: Intense scan
Scan
Cancel
Command: nmap -T4 -A -v 192.168.2.*
Hosts
Services
OS Host
192.168.2.1
192.168.2.100
192.168.2.101
192.168.2.103
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Port Protocol State Service Version
135 tcp open msrpc Microsoft Windows RPC
139 tcp open netbios-ssn Microsoft Windows netbios-ssn
445 tcp open microsoft-ds Microsoft Windows Server 2008 R2 - 201
3389 tcp open ms-wbt-server Microsoft Terminal Service
49152 tcp open msrpc Microsoft Windows RPC
49153 tcp open msrpc Microsoft Windows RPC
49154 tcp open msrpc Microsoft Windows RPC
49155 tcp open msrpc Microsoft Windows RPC
49161 tcp open msrpc Microsoft Windows RPC
49176 tcp open msrpc Microsoft Windows RPC
4:29 PM
7/14/2018

INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***10 POINTS***

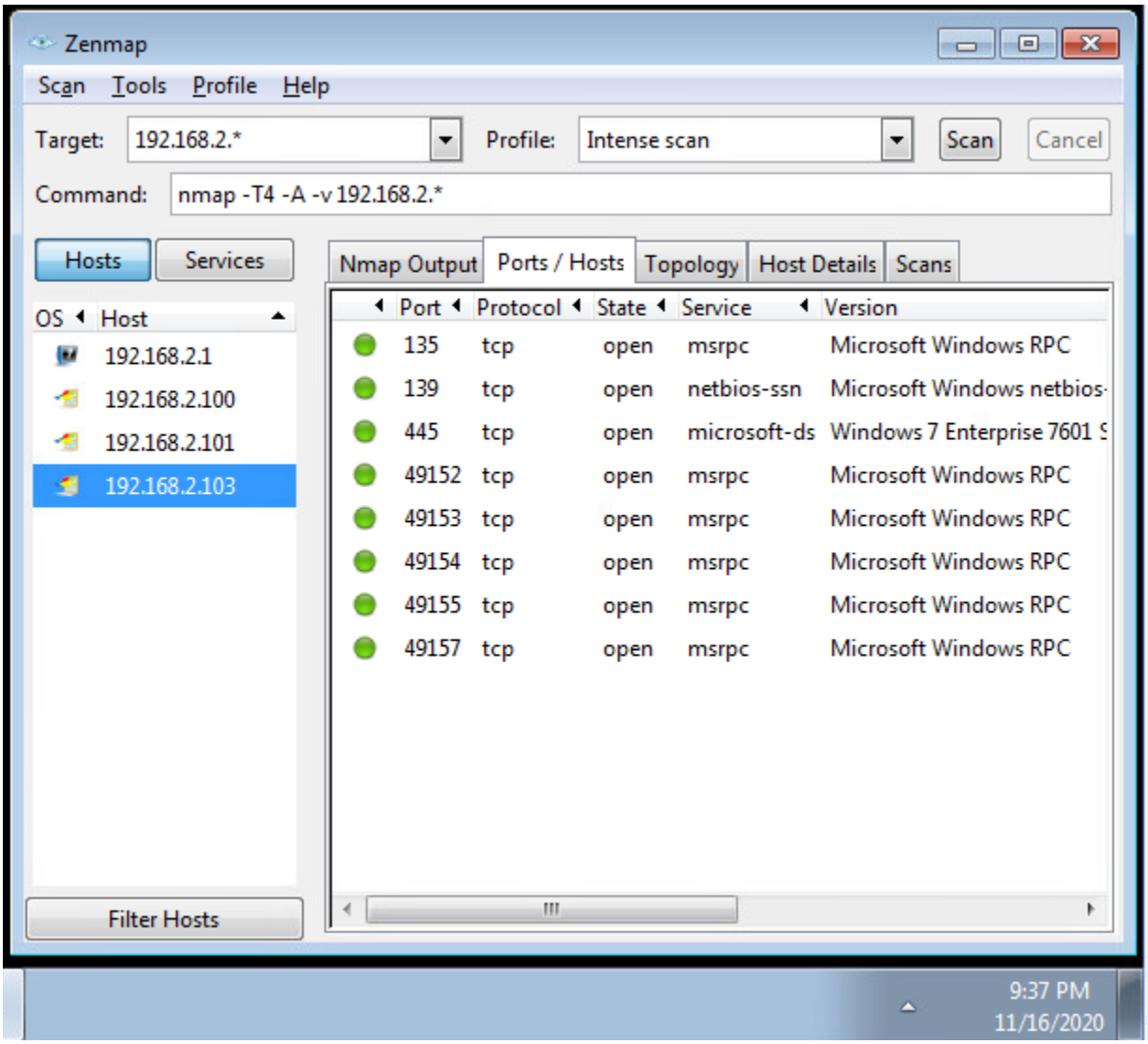


- Click on **192.168.1.103** in the left pane, to select the **CNSLabWin7** system
- Click on " **Port / Hosts** " folder tab
- The right pane changes to show port information about **CNSLabWin7**
- Move the window to the lower right corner in order to take a screenshot that includes the date/time
- Using Snipping Tool, take a screenshot. It should look like



WARNING: DO NOT CLOSE **Zenmap** until you have answered the following questions below

INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***10 POINTS***



- This QUESTION is worth ***5 points***. Write your answer in the box.

What is the MAC address for **192.168.2.100** ? HINT: Look at the **Host Details** output for **192.168.2.100**

The MAC address of 192.168.2.100 is 00:50:56:01:19:FB

- This QUESTION is worth ***5 points***. Write your answer in the box.

How many ports are scanned on **192.168.2.103** ? HINT: Look at the **Nmap** output

nmap scanned 1000 ports on 192.168.2.103 to realize the few that were open.

- This QUESTION is worth ***5 points***. Write your answer in the box.

Is TCP Port **555** open on **192.168.1.101** ? HINT: Look at the **Port / Hosts** output for **192.168.2.103**

Port 555 does not seem to be an open port on 192.168.2.101

- This QUESTION is worth ***5 points***. Write your answer in the box.

How many total services were scanned on the 3 hosts ? HINT: Look at the **Nmap** output and search for "Scanning xx services on 3 hosts"

After finding all three hosts, nmap scanned 26 services on all three hosts. “Scanning 26 services on 3 hosts”

- This QUESTION is worth ***5 points***. Write your answer in the box.

What **Operating System** is running on **192.168.2.100** ? HINT: Look at the **Host Details** output for **192.168.2.100**.

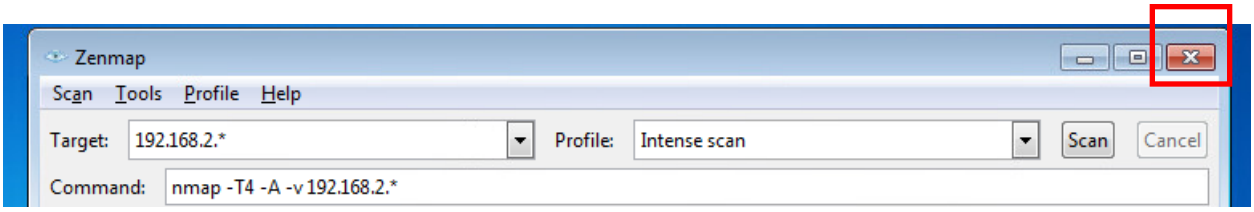
According to nmap, Windows Server 2012 is running on 192.168.2.100

- This QUESTION is worth ***5 points***. Write your answer in the box.

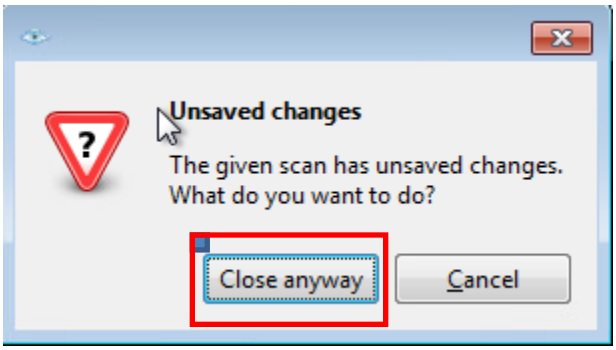
Is the **ldap** service running on **192.168.2.101** ? HINT: Look at the **Port / Hosts** output for **192.168.2.101**

ldap does not seem to be running on 192.168.2.101.
I dap does seem to be running, however, on 192.168.2.100 on port 389 and port 3268.

- Click on the **X** to close **Zenmap**



- The following window appears
- Click on **Close Anyway**



THIS IS THE LAST PAGE OF THE LAB